

Precautionary Taxation*

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Abstract

Using effective tax rates constructed from revenue statistics and national accounts for 38 countries, which include both emerging and more advanced economies, we document substantial dispersion in effective capital tax rates across countries, ranging from 4.9% to 53%, with a coefficient of variation of 0.51 (compared to 0.42 for the labor tax). We then establish three stylized facts: i) a negative relation between aggregate volatility and capital taxes, ii) this relation becomes stronger in countries with better institutions, and iii) the relation is absent in the case of the labor tax. These facts survive when one instruments output volatility with the growth volatility of a country's trading partners.

Existing models cannot account for these facts because they abstract either from politics or from risk. We thus develop a general equilibrium model in which a self-interested politician sets taxes under aggregate productivity risk. The model predicts that the capital tax falls with aggregate risk because of a precautionary motive, and that the effect is stronger when political frictions are smaller. We calibrate the model to two reference economies—the United States for the advanced world and Chile for the emerging one. In the US, aggregate risk lowers the capital tax by 21 percentage points, whereas in Chile the same risk lowers it by less than a tenth of a percentage point. Moreover, the labor tax is barely affected by aggregate risk in both economies.

Keywords: capital taxation, aggregate risk, political frictions, limited commitment, precautionary motive, emerging economies.

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1 Introduction

What determines how heavily a country taxes capital? The theory of optimal taxation prescribes a long-run capital tax close to zero (Judd, 1985; Chamley, 1986), while the political economy literature explains why governments that cannot commit sustain a positive one (Acemoglu et al., 2008). However, neither approach explains why capital taxes differ so much across countries. We argue that these differences are related to a dimension that the literature has overlooked: the aggregate risk an economy faces. Under aggregate risk, governments have a precautionary reason to lower the capital tax, since a lighter capital tax helps the economy smooth consumption. This precautionary motive operates only where political frictions sustain a positive capital tax in the first place. Emerging economies face the most aggregate risk, which makes them the natural place to ask how far the motive reaches.

This paper examines how aggregate risk shapes capital taxation across countries. The main empirical finding is that output volatility lowers the effective capital tax: an increase in volatility of one percentage point reduces the capital tax by about six percentage points, the effect is stronger in countries with better institutions, and there is no effect on the labor tax. To identify this effect, we instrument volatility with the trade-weighted volatility of a country's export partners. To rationalize these facts, we develop a general equilibrium model in which a self-interested politician sets taxes under aggregate risk, and calibrate it to two reference economies: the United States for the advanced world and Chile for the emerging one. In the United States the model replicates the three facts: aggregate risk lowers the capital tax by 21 percentage points, the reduction is larger where institutions are stronger, and the labor tax does not respond. In Chile the same risk lowers the capital tax by less than a tenth of a percentage point. These results suggest that aggregate risk is a first-order determinant of capital taxation, and that it operates through political frictions.

The negative relation between volatility and capital taxation is evident in the raw data. Figure 1 displays country-average effective tax rates against output volatility for our sample: the capital tax declines steeply with volatility, while the labor tax shows no relation. The effective rates are constructed from national accounts and measure the taxes economies actually collect; Section 2 describes the data and presents the regressions.

Effective tax rates and output volatility, country averages

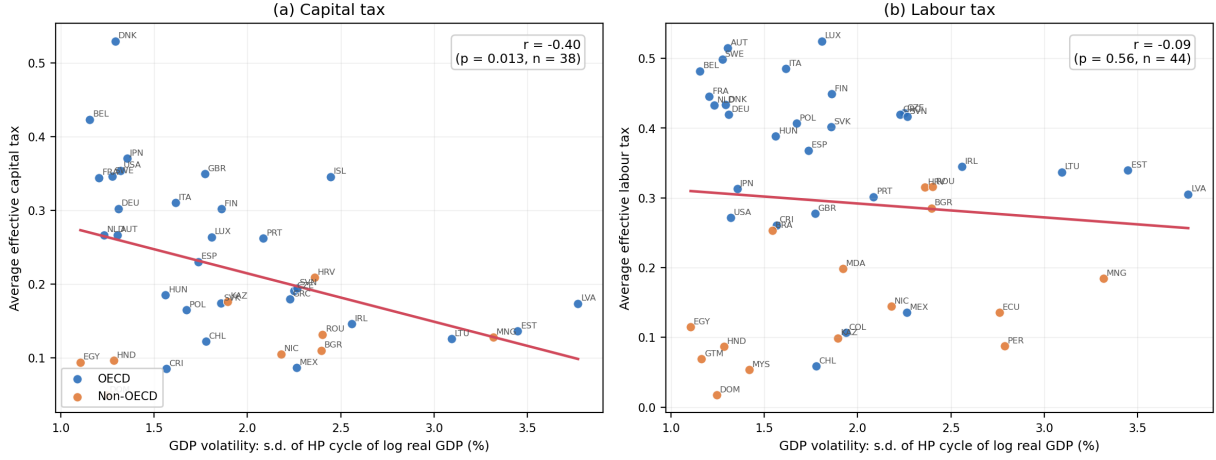


Figure 1: Effective tax rates and output volatility. Country averages of the effective capital tax (left panel) and the effective labor tax (right panel) against the volatility of output; each point is a country. The capital tax declines with volatility; the labor tax does not. Section 2 describes the data and measurement.

Existing models cannot account for these facts because they abstract either from politics or from risk. When governments can commit, the optimal capital tax is zero in the long run (Judd, 1985; Chamley, 1986; Diamond and Mirrlees, 1971) and there is nothing for volatility to reduce. When aggregate risk is added to this framework, state-contingent policy absorbs the shocks with essentially no effect on the average capital tax (Zhu, 1992; Chari et al., 1994; Aiyagari et al., 2002). When idiosyncratic risk is added to this framework, Aiyagari (1995) showed that the optimal policy wants to tax capital because the laissez-faire economy accumulates too much capital, implying a positive relation between risk and the capital tax in this case. In turn, the political economy models that generate a positive capital tax are deterministic: the politician of Acemoglu et al. (2008, 2011) faces no aggregate shocks, and the stochastic economy of Yared (2010) has no capital. Our model combines the two ingredients: a positive capital tax sustained by political frictions and aggregate risk.

In the model, a representative household accumulates capital and supplies labor in a neo-classical growth economy with aggregate productivity risk. The government is run by a self-interested politician who sets taxes and diverts public resources, disciplined only by the citizens' ability to replace him; the quality of institutions determines the share of output a deviating politician can seize. We show that the best sustainable equilibrium of this economy solves a planning problem, as in its deterministic predecessors. In this problem, the capital tax is an expected wedge with two components: a political component, proportional to the

tightness of the politician’s constraint, and an implementability component that reflects the standard cost of financing government spending with distortionary taxes.

The model delivers three analytical results. First, without aggregate risk the political constraint binds and sustains a capital tax that is predominantly political and vanishes as institutions become perfect. Second, under aggregate risk this characterization no longer holds—the friction survives only on average—and risk introduces a covariance between the household’s marginal utility and the return to capital that reduces the capital tax in every state. Third, institutions enter the capital tax only through its political component, so risk and institutions can interact only through the political side of the tax. These results leave undetermined the balance between two opposite effects on the level of the capital tax: the political friction pushes the tax up, while the precautionary motive pushes it down. The quantitative model allows us to disentangle the two effects.

The main quantitative result is that aggregate risk lowers the capital tax by 21 percentage points in the United States. We calibrate the model separately to each reference economy, fixing the political discount factor from data on leader tenure and choosing the remaining parameters to match the capital-output ratio, output volatility, the capital tax, and the size of government. Without risk the United States economy sustains a capital tax of 57 percent; at its own volatility the tax is 36 percent, and raising volatility to 2.1 percent, above three fifths of the countries in our sample, lowers it to 25 percent. Sweeping the quality of institutions at the calibrated volatility, the reduction ranges from -1 percentage points where institutions are weak to 22 points where they are strong.

The emerging economy behaves differently, and the difference is the paper’s second quantitative result. Chile sustains a capital tax of 12 percent without risk, and at its own volatility, which is higher than the United States’, the tax is 12 percent. Across the same sweeps of volatility and institutions the capital tax never moves by as much as one percentage point, while the buffer stock of capital that risk induces is as large as in the United States. In the data, the capital taxes of countries with weak institutions show the same absence of a response to volatility.

The main intuition behind our results is that aggregate risk makes capital a poor hedge. Capital pays off in good states, when the household’s marginal utility is low, so risk lowers the value of the claim that the capital tax prices, and the tax that supports the politician’s wedge falls. In this setting of Ramsey taxation with political constraints, the precautionary motive operates on the political component of the tax. Without the politician, the model produces neither a sizable capital tax nor a response of the capital tax to risk: through the

lens of the model, the volatility patterns in the data are evidence of political frictions.

We contribute to two literatures. To the macro political economy literature, we develop a model with aggregate risk and political frictions and show that risk plays a first-order role in explaining the capital tax. To the literature on cross-country differences in fiscal policy, we provide causal evidence on volatility and capital taxation and a model that rationalizes it.

Our first contribution is to the macro political economy literature on taxation under limited commitment (Thomas and Worrall, 1988; Kocherlakota, 1996; Acemoglu et al., 2008; Yared, 2010). In this literature, political frictions explain why the capital tax is positive, but aggregate risk plays no role. We develop a model with both ingredients, re-establish the characterization of the best sustainable equilibrium in the combined environment, and show that aggregate risk is a first-order determinant of the capital tax that the political friction sustains.

Our second contribution is to the literature that studies cross-country differences in fiscal policy. Some of these papers focus on the size of government (Galí, 1994; Rodrik, 1998; Janiak and Santos Monteiro, 2016), others on the cyclical behavior of fiscal policy (Fatás and Mihov, 2001; Bauducco and Caprioli, 2014); the structure of taxation has received less attention. We provide empirical evidence that volatility lowers the capital tax, and our model shows how aggregate risk helps to understand cross-country differences in capital taxation. Because emerging economies are the part of the cross-section where aggregate risk is largest (Acemoglu et al., 2003; Aguiar et al., 2009), we calibrate the model to one of them and report what the mechanism delivers there.

The two papers closest to ours are Acemoglu et al. (2011) and Yared (2010). Relative to the first, we add aggregate risk; relative to the second, we add capital, since in his economy output is produced with labor only. The combination is what generates our results: with risk and capital together, a precautionary channel shapes the capital tax, and this channel cannot operate in either of these models—without risk there is nothing to insure against, and without capital there is no capital tax to adjust. We show that this channel is quantitatively central.

The paper proceeds as follows. Section 2 documents the three facts. Section 3 develops the model and derives the tax formulas. Section 4 presents the comparative statics results. Section 5 calibrates the model to the two reference economies and presents the quantitative results. Section 6 concludes.

2 Motivating evidence

2.1 Data

Our analysis relates country-average tax rates to country-level volatility, so the unit of observation is a country. We assemble a country-year panel of effective tax rates, real output, and institutional indicators, and collapse it to country means over the years available for each country. The underlying series were retrieved programmatically from the data services of the OECD, the World Bank, the United Nations and V-Dem, and the tax rates were computed at the country-year level before averaging. Appendix A lists, for every country in the sample, the years averaged and the source of each input. The capital-tax regressions use 38 countries and the labor-tax regressions the 37 of them for which the labor tax is available.¹

The tax rates are realized average effective rates constructed from revenue statistics and national accounts, following Mendoza et al. (1994). The construction has to solve one apportionment problem: revenue statistics report personal income tax collections as a single total, without splitting them between labor and capital income. The solution of Mendoza et al. (1994) is to compute the household income-tax rate on the household's total pre-tax income,

$$\tau_h = \frac{PIT}{OSPUE + PEI + W}, \quad (1)$$

where PIT is personal income tax revenue (heading 1100 of the OECD Revenue Statistics), $OSPUE$ is the operating surplus of private unincorporated enterprises (i.e., the mixed income of the self-employed), PEI is the property income households receive (i.e., dividends, interest, and rents), and W is wages and salaries received by households. The rate τ_h is then applied to the labor part of household income in the labor tax and to the capital part in the capital tax.

The labor tax collects every levy on labor income and divides it by the economy's labor cost,

$$\tau_\ell = \frac{\tau_h W + SSC + PT}{W + SSC_{emp}}, \quad (2)$$

where SSC is total social security contributions (heading 2000) and PT is payroll taxes (heading 3000). The denominator in equation (2) adds the employer part of contributions SSC_{emp} (heading 2200) to wages and salaries, implying that the base is total labor cost. The capital tax collects the levies on capital income and divides them by the economy's capital

¹Iceland has the net capital rate but is missing the labor rate.

income,

$$\tau_k = \frac{\tau_h (OSPUE + PEI) + CIT + PROP + FIN}{OS}, \quad (3)$$

where *CIT* is corporate income tax revenue (heading 1200), *PROP* is recurrent taxes on immovable property (heading 4100), *FIN* is taxes on financial and capital transactions (heading 4400), and the base *OS* is the operating surplus of the total economy, including mixed income, net of the consumption of fixed capital. The net base is the choice that matters for comparability with the theory: the model’s capital tax falls on the return net of depreciation (Section 3), so measurement and model tax the same object.²

The consumption tax is used only to construct the labor wedge in the robustness analysis. It is revenue from general taxes on goods and services and from excises (headings 5110 and 5121) divided by final consumption expenditure valued before those taxes, $\tau_c = (GST + EXC)/(C + G - GW - GST - EXC)$, where *C* and *G* are private and government final consumption and *GW* is government compensation of employees. Because all rates are computed from realized collections, they measure the taxes economies actually pay rather than statutory schedules.

Output volatility is the standard deviation of the cyclical component of log real GDP. For each country we take the longest uninterrupted run of annual observations, requiring at least ten consecutive years, filter the log series with the Hodrick–Prescott filter with the smoothing parameter of Ravn and Uhlig (2002) for annual data, and express the resulting standard deviation in percent.³

Institutional quality is a composite of thirteen indices: six Worldwide Governance Indicators—control of corruption, government effectiveness, political stability, rule of law, regulatory quality, and voice and accountability (Kaufmann et al., 2011)—and seven Varieties of Democracy indices—political corruption, rule of law, judicial constraints, legislative constraints, electoral democracy, liberal democracy, and the liberal component (Coppedge et al., 2026). Each index is normalized to the unit interval, the corruption index is reversed so that higher values always indicate better institutions, and each index is averaged over 1996 onward within each country. The thirteen country means are then averaged, giving one value per country. We average them because the thirteen indices are highly collinear, but the robustness analysis re-runs the main specification with each index entered separately. The

²For overlapping country-years the constructed rates match the values published by Mendoza et al. (1994) to within about one and a half percentage points.

³Real GDP is from OECD national accounts for OECD members and from the United Nations national-accounts database otherwise. Volatility is a within-country moment, so differences in levels or units across sources do not affect it.

Table 1: Summary statistics

	Mean	Std. dev.	Min	Max	Countries
Capital tax	0.219	0.111	0.049	0.530	38
Labor tax	0.325	0.138	0.017	0.525	37
Output volatility (%)	1.928	0.674	1.105	3.772	38
Institutional quality	0.741	0.176	0.298	0.928	38
Tax revenue / GDP (%)	20.9	6.2	9.5	39.9	38
Debt / GDP (%)	57.4	30.3	9.5	149.1	38

Notes. Country means over the capital-tax regression sample; the labor tax over the 37 of these countries for which it is available. Tax revenue and debt are in percent of GDP.

governance indicators begin in 1996 and we use the same window for the V-Dem indices, so the composite reflects the recent institutional environment, which is appropriate for a cross-section of long-run average tax rates.

Two further variables enter the regressions. Fiscal capacity is measured by total tax revenue in percent of GDP, from the state-capacity dataset of Hanson and Sigman (2021), averaged by country, and an OECD indicator is included throughout. The robustness analysis also controls for public indebtedness, measured by general government gross debt in percent of GDP from the IMF World Economic Outlook over 1980–2024, averaged by country. The instrument for volatility is built from trading partners’ growth and is described together with the identification strategy below. Table 1 reports summary statistics for the regression sample: the dispersion the regressions exploit is large, with capital taxes ranging from 5 to 53 percent of the net capital base and output volatility from 1.1 to 3.8 percent.

2.2 Taxation and aggregate volatility

In Table 2, we show some correlations between taxation—both capital and labor—and output volatility using OLS regressions. In the first four columns, the dependent variable is the average effective capital tax rate, while the last four columns show regressions for the average effective labor tax rate. All regressions consider HC1 robust standard errors (reported in parentheses) to assess the significance of the estimates. The first column reports regression results for the capital tax on a measure of output volatility for a cross section of 38 countries. 29 of these countries are OECD members and 9 are non-OECD.⁴ Volatility is the output-volatility measure described in the Data subsection. Columns (1) and (5) are the regression

⁴The 9 non-OECD countries are Bulgaria, Croatia, Dominican Republic, Egypt, Honduras, Kazakhstan, Mongolia, Nicaragua and Romania.

Table 2: Average effective tax rates and output volatility: OLS regressions

	Capital tax				Labor tax			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Output volatility	−0.060*** (0.019)	−0.061*** (0.017)	−0.060*** (0.016)	−0.052*** (0.014)	−0.010 (0.018)	−0.012 (0.014)	−0.010 (0.014)	−0.001 (0.013)
Institutional quality		0.469*** (0.117)	0.388*** (0.113)	0.109 (0.162)		0.558*** (0.109)	0.495*** (0.132)	0.203 (0.167)
Quality × Volatility			−0.263*** (0.092)	−0.234*** (0.064)			−0.205* (0.115)	−0.155 (0.095)
Tax revenue / GDP				0.008** (0.003)				0.009** (0.004)
OECD dummy	0.120*** (0.031)	−0.039 (0.035)	−0.024 (0.038)	0.025 (0.046)	0.198*** (0.043)	0.010 (0.045)	0.022 (0.054)	0.074 (0.051)
R^2	0.375	0.548	0.596	0.707	0.400	0.562	0.581	0.664
Countries	38	38	38	38	37	37	37	37

Notes. OLS on country means; HC1 robust standard errors in parentheses. Tax revenue / GDP is total tax revenue in percent of GDP (Hanson and Sigman, 2021), averaged by country. Columns (5)–(8) use the 37 countries of the capital-tax sample for which the labor tax is available. Columns (3), (4), (7) and (8): quality and volatility mean-centered. ***, **, *: 1%, 5%, 10%.

counterparts of Figure 1 in the introduction, with the difference that they control for an OECD dummy variable (as do all regressions in the table). Column (1) confirms that the correlation between output volatility and the capital tax rate is significant. This is the first statistical fact we wish to put forward in this paper.

Emerging countries with weaker institutions tend to have more volatile output (Acemoglu et al., 2003; Aguiar et al., 2009). It could thus be the case that the correlation between the tax rate and volatility is spurious and simply reflects the role played by institutional quality. For this reason, column (2) controls for the composite institutional quality index described in the Data subsection, with higher values meaning stronger institutions.⁵ While column (2) confirms that countries with better institutions are characterized by higher tax rates, it also shows that controlling for this does not reduce the significance (nor the size) of the coefficient associated with output volatility.

In column (3), we explore the second fact we wish to put forward in this paper: the relation between capital tax and output volatility gets stronger in countries with stronger institutions. For this reason, we interact our composite index on institutional quality with our measure of output volatility.⁶ The interaction turns out significant at the 1% level, confirming that the relation becomes stronger when institutional quality is higher, while the volatility coefficient,

⁵Below we include a robustness analysis where we re-run the specification with each of the 13 indices individually to show the result is not driven by any single measure.

⁶Because quality and volatility are mean-centered in columns (3), (4), (7) and (8), the volatility coefficient is the effect evaluated at the sample-average institutional quality, and the interaction gives how that slope steepens with institutions.

now evaluated at the average level of institutional quality, is essentially unchanged.

Two issues may affect the results in column (3). First, countries with weaker institutions suffer from lower fiscal capacity (Besley and Persson, 2009). It could be the case that the positive correlation between institutional quality and the capital tax is due to the fact that countries with better institutions levy more taxes. Second, the literature on fiscal policy has also shown that countries with large governments tend to display lower output volatility (Galí, 1994; Fatás and Mihov, 2001; Janiak and Santos Monteiro, 2016). Hence, column (4) adds tax revenue as a share of GDP to the list of regressors to address these issues. It enters positively and significantly. Its inclusion absorbs the effect of institutional quality, which is no longer significant, but it leaves the coefficient on output volatility essentially unchanged, at -0.052 , still significant at the 1% level. The interaction coefficient between output volatility and institutional quality and its significance do not change either.

Finally, in columns (5)-(8), we reproduce the regressions from columns (1)-(4) changing the dependent variable for the average effective labor tax rate.⁷ The results are similar with the difference that output volatility is never significant in these regressions. This is the third fact we wish to put forward: there is no strong association between output volatility and the labor tax. The interaction with institutional quality is significant at the 10% level in column (7), but it becomes insignificant once fiscal capacity is added in column (8).

The results in Table 2 display significant statistical correlations between output volatility and capital tax. However, it is hard to make any causal claim from them because output volatility is an endogenous variable. The literature on government size indeed suggests that taxes (or transfers, which are correlated with taxes) may act as automatic stabilizers, lowering aggregate volatility (Galí, 1994; Rodrik, 1998; Fatás and Mihov, 2001; Janiak and Santos Monteiro, 2016). To address the concern that output volatility is itself a function of fiscal policy, we instrument it with a measure of external business-cycle volatility that a country's own tax policy cannot influence. Specifically, our identification strategy follows Badinger (2010), who instruments domestic output volatility with exogenous volatility spillovers from abroad, in line with well-documented evidence that trading partners' fluctuations are transmitted into domestic volatility (di Giovanni and Levchenko, 2009; Caselli et al., 2020). The instrument is a trade-weighted average of a country's trading partners'

⁷The labor-tax regressions use the 37 countries of the capital-tax sample for which the labor tax is available: Iceland has the net capital rate but is missing the labor rate. Seven further economies report labor-tax inputs but not consumption of fixed capital, so their net capital rate cannot be computed (only the gross one exists): Brazil, Colombia, Ecuador, Guatemala, Moldova, Malaysia, and Peru. The robustness analysis below adds these seven economies to the labor-tax sample.

Table 3: IV estimates: effective tax rates and output volatility

	Capital tax				Labor tax			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Output volatility (2SLS)	-0.045** (0.020)	-0.051*** (0.017)	-0.044*** (0.016)	-0.046*** (0.014)	0.011 (0.027)	0.005 (0.020)	0.010 (0.019)	0.007 (0.015)
Institutional quality		0.468*** (0.111)	0.361*** (0.126)	0.098 (0.157)		0.557*** (0.107)	0.485*** (0.126)	0.200 (0.143)
Quality $\times$ Volatility (2SLS)			-0.342*** (0.128)	-0.258*** (0.093)			-0.230 (0.163)	-0.139 (0.135)
Tax revenue / GDP				0.008*** (0.003)				0.009** (0.004)
OECD dummy	0.122*** (0.027)	-0.037 (0.033)	-0.017 (0.042)	0.028 (0.046)	0.201*** (0.038)	0.013 (0.044)	0.027 (0.053)	0.076 (0.047)
First-stage F (volatility)	96.7	97.8	102.0	102.4	99.4	100.3	105.2	107.1
First-stage F (interaction)	—	—	91.9	95.4	—	—	96.7	95.9
Countries	38	38	38	38	37	37	37	37

Notes. Two-stage least squares on country means; robust standard errors in parentheses. Columns (5)–(8) use the 37 countries of the capital-tax sample for which the labor tax is available. Output volatility is instrumented by trade-weighted partner-growth volatility and, in columns (3), (4), (7) and (8), the interaction by its product with institutional quality; quality and volatility are mean-centered in these columns. First-stage F is the robust first-stage statistic when only volatility is instrumented and the Sanderson–Windmeijer conditional statistic for each endogenous regressor otherwise. Tax revenue / GDP as in Table 2. ***, **, *: 1%, 5%, 10%.

output volatility. Formally, for each country i we first build a foreign-demand growth series,

$$g_{i,t}^f = \sum_{j \neq i} \omega_{ij} g_{j,t},$$

where $g_{j,t}$ is partner j 's real GDP growth in year t and ω_{ij} is the average share of i 's merchandise exports going to j , held fixed at its 2000-2019 mean. The instrument is thus the standard deviation of $g_{i,t}^f$ over 1985-2019. Bilateral export weights are from the IMF Direction of Trade Statistics and partner growth rates from the World Development Indicators. By construction the instrument is built entirely from foreign output and excludes country i 's own GDP.⁸

Table 3 reproduces the results of Table 2 using the IV strategy. Two properties make our instrument credible. First, it is strongly relevant: open economies import their trading partners' cycles, so countries that trade with volatile neighbours are themselves volatile. Partner-growth volatility is correlated 0.79 with realised output volatility, and the first-stage F -statistics displayed in Table 3 exceed 90 throughout, far above conventional weak-instrument thresholds (Stock and Yogo, 2005). In the specifications that also instrument the

⁸We also tried to instrument volatility with terms of trade. Unfortunately, this instrument did not work in the sense that the F statistic of the first stage was too low. The reason is because we do not have many small open economies in our sample.

quality-volatility interaction we report the Sanderson and Windmeijer (2016) conditional first-stage statistic for each endogenous regressor. Second, it is plausibly excludable: a country’s capital-tax policy does not move its trading partners’ business cycles, so foreign volatility can affect the home capital tax only by transmitting to home output volatility. The main threat to the exclusion restriction is that partner volatility is correlated with region and development, in the sense that economies in volatile neighborhoods are disproportionately emerging and transition markets. It could potentially proxy for unobserved correlates of taxation rather than for output volatility as such. We mitigate this in some ways as the OECD dummy is included throughout and the institutional-quality index in columns (2)-(4).

Comparing Table 3 with the capital tax regressions in Table 2 shows similar results. The IV estimates of the volatility coefficient are slightly smaller in absolute value than their OLS counterparts, between -0.044 and -0.051 across the four columns, and remain significant at the 1% level in columns (2)-(4), while the interaction coefficient is, if anything, larger. Columns (5)-(8) confirm that the labor tax is not related to volatility also under the IV strategy. The relationship does not seem to be an artifact of the reverse causality that the automatic-stabilizer literature would predict. A related threat to the exclusion restriction is the compensation channel of Rodrik (1998): economies exposed to greater external risk run larger governments to provide social insurance, and larger governments could require higher taxes. The first part of the channel is present in our data since the ratio of government consumption to GDP rises with our instrument. However, the second part is not, in the sense that tax revenue as a share of GDP is unrelated to the instrument, and column (4) of Table 3 controls for it directly.⁹ The larger governments of economies facing more external risk are thus not financed by higher capital taxes.

The interaction between volatility and institutional quality thus appears significant both for the OLS and the IV estimations. In Figure 2, we rely on an alternative way to illustrate that the impact of volatility on the capital tax varies with institutional quality. In particular, we divide the sample between the countries with institutional quality above the median and those below it, and estimate the two-stage least squares slope of the capital tax on output volatility within each subsample separately. Both subsamples have strong first stages ($F = 90$ and 78). The graph shows that the negative relationship is significant only among strong-institution countries, while it is not among weak-institution countries.¹⁰ If one does

⁹A regression of government consumption as a share of GDP on our instrument, controlling for institutions and OECD status, yields a positive coefficient significant at the 1% level. The same regression for tax revenue as a share of GDP has a p-value of 0.98.

¹⁰A possible explanation for the negative relation between output volatility and the capital tax turns on

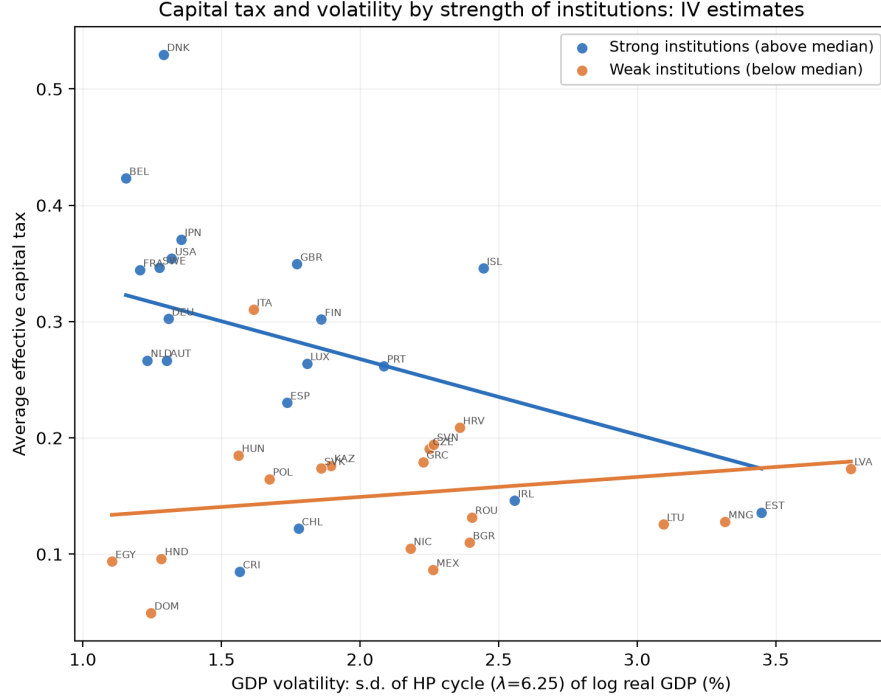


Figure 2: Capital tax and volatility by strength of institutions: IV estimates. Countries are split at the median of the institutional-quality index. Each line is the within-group two-stage least squares slope of the average effective capital tax on output volatility, instrumented by trade-weighted partner-growth volatility (first-stage $F = 90$ and 78 for the strong- and weak-institution groups). The negative relationship is present only among strong-institution countries (-0.065 , $p = 0.03$) and absent among weak-institution countries ($+0.017$, $p = 0.37$).

the same exercise in the case of the labor tax, there is no significant impact in the case of the strong-institution group: the instrument is strong ($F = 79$), but the labour-tax slope on instrumented volatility is an insignificant 0.008 , with a p-value of 0.82 .

2.3 Robustness

We now investigate the robustness of the three facts we described previously: 1) there is a significant negative relation between output volatility and the capital tax rate, 2) this relation

measurement. The Mendoza et al. (1994) capital tax rate divides revenue by net operating surplus. The denominator is inflated by informality and self-employment income, precisely where enforcement is weak. This should push measured capital taxes down in the volatile economies, which are also economies with weak institutions. The distinctive implication of this story is that the negative relationship between volatility and the capital tax should be concentrated among weak-institution countries. The data shows the opposite since the slope in Figure 2 suggests it is a strong-institution phenomenon and it is flat precisely where informality concerns are most severe.

Table 4: Capital-tax interaction across individual institutional indices: IV estimates

Institutional index (used singly)	$\beta_{\text{volatility}}$	$\beta_{\text{quality} \times \text{vol.}}$
WGI Control of corruption	-0.049***	-0.227***
WGI Government effectiveness	-0.047***	-0.279***
WGI Political stability	-0.041***	-0.685***
WGI Rule of law	-0.048***	-0.296***
WGI Regulatory quality	-0.046***	-0.319***
WGI Voice & accountability	-0.053***	-0.419***
V-Dem Political corruption (rev.)	-0.048***	-0.103***
V-Dem Rule of law	-0.044***	-0.139**
V-Dem Judicial constraints	-0.031**	-0.239**
V-Dem Legislative constraints	-0.042***	-0.242***
V-Dem Electoral democracy	-0.043***	-0.233**
V-Dem Liberal democracy	-0.044***	-0.196**
V-Dem Liberal component	-0.036**	-0.304***

Notes. Each row is a separate two-stage least squares regression of the average capital tax on the named institutional index, output volatility, their interaction, tax revenue as a share of GDP, and the OECD dummy, as in column (4) of Table 3; output volatility and its interaction with the index are instrumented by trade-weighted partner-growth volatility and its interaction with the index (index and volatility mean-centered; each index averaged over 1996 onward; robust SE; $N = 38$). The V-Dem political-corruption index is entered reverse-coded (rev.), so that, like the others, higher values denote better institutions. Only the two volatility-related coefficients are shown. ***, **, and * denote significance at the 1%, 5%, and 10% level.

is stronger for countries with better institutional quality, 3) the relation is not significant in the case of the labor tax.

The first concern we have is the variable for institutional quality we use. The one used in Tables 2 and 3 is a composite indicator that relies on six measures from the World Bank Worldwide Governance Indicators and seven V-Dem indices (Varieties of Democracy). All these indicators are collinear: this is the reason why we chose to work with a composite indicator. In Table 4, we show the robustness of the results in column (4) of Table 3 by using each one of the indicators separately instead of the composite indicator. All 13 regressions suggest that volatility lowers the capital tax, and more so where institutions are stronger: every volatility and interaction coefficient is significant at the 5% level or better.¹¹

A second concern is that the significant interaction that has been estimated in Tables 2

¹¹In the case of Political corruption from V-Dem, we transformed the variable into 1 minus the index so that a higher value indicates better institution, inline with the rest of the indicators.

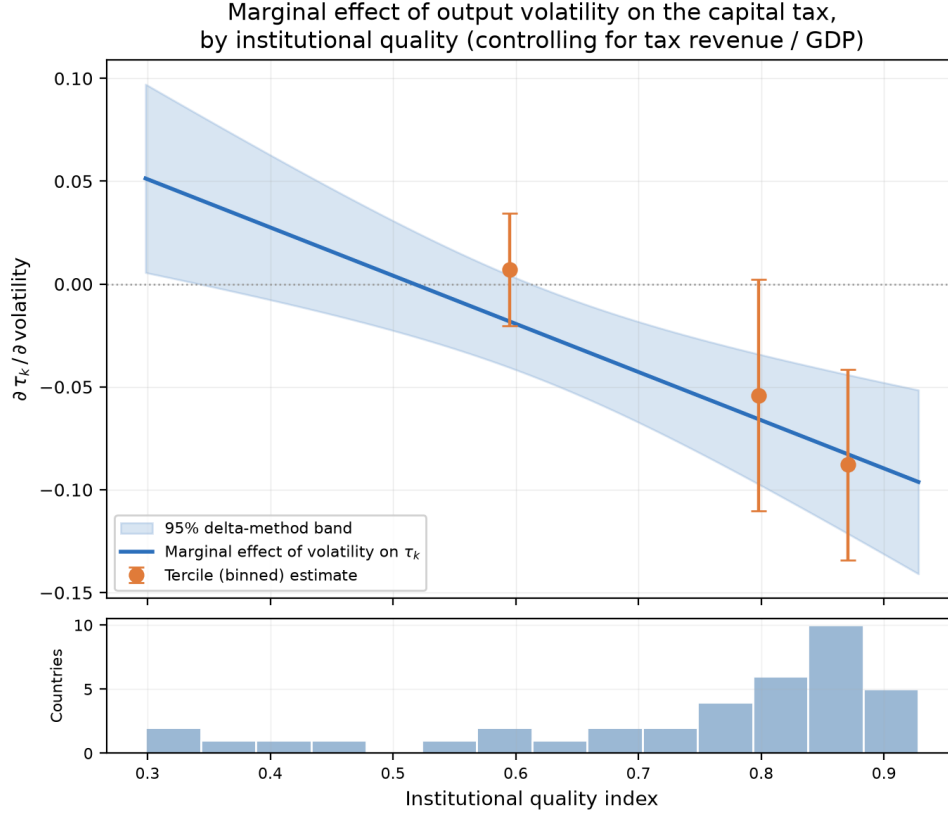


Figure 3: Marginal effect of output volatility on the capital tax, by institutional quality, controlling for tax revenue. The solid line and shaded band are the point estimate and 95% delta-method confidence band from the OLS interaction model (capital tax on volatility, institutional quality, their interaction, the OECD dummy, and tax revenue as a share of GDP); orange points are volatility slopes estimated within quality terciles (the binning diagnostic of Hainmueller et al. (2019)), holding tax revenue and the OECD dummy constant, plotted at tercile medians with 95% intervals. The lower panel shows the distribution of the institutional-quality index across the 38 sample countries.

and 3 is an artifact produced by the functional form we impose on the equation to be estimated: we force the marginal effect of volatility to vary linearly with institutional quality. The interaction turns out significant in the regressions but it can be because the correct relation between capital tax and volatility is non linear. Surveying a large body of published interaction estimates, Hainmueller et al. (2019) in the political science field find that such failures are widespread: many published interactions flatten or reverse once linearity is relaxed.

Figure 3 reproduces the analyses of Hainmueller et al. (2019) in the case of the OLS regression of column (4) of Table 2. We split the sample into low-, middle- and high-institutional quality bins and estimate the slope of interest separately within each bin (by interacting volatility

with bin indicators). Figure 3 places the three estimates at the within-bin medians of the institutional quality indicator, above a histogram showing where the institutional quality’s mass actually lies. The diagnostic can be made from the Figure as follows. If the linear interaction describes the data, the binned slopes align with the straight line it implies. If it is spurious, they do not.¹²

In our case the binned slopes line up with the linear prediction: the volatility slope falls monotonically from a statistically insignificant 0.01 in the low-quality bin to -0.05 in the middle bin and a significant -0.09 in the high-quality bin. The middle bin is the most informative for this check because a straight line is largely pinned down by its endpoints. Any curvature in the relationship would show up there as a departure from the line. But the estimate falls squarely on the linear prediction. We conclude that the significant interaction is not an artifact of the imposed functional form, but reflects a genuine and approximately linear strengthening of the volatility effect as institutional quality rises.¹³

A third robustness exercise we implement is motivated by the fact that the distortion generated by the labor tax rate in the Neoclassical growth model is actually a consequence of both the labor tax and the consumption tax. Hence, one could argue that the relevant tax to look at is a combination of both labor and consumption taxes. For this reason, we redo the OLS labor tax regressions of columns (5)-(8) of Table 2 in Table 5 by considering the labor wedge instead of the labor tax as a dependent variable.¹⁴ Labor wedge in country i is defined as

$$\tilde{\tau}_{\ell,i} = 1 - \frac{1 - \tau_{\ell,i}}{1 + \tau_{c,i}},$$

where $\tilde{\tau}_{\ell,i}$ is the labor wedge, $\tau_{\ell,i}$ is the average effective labor tax rate and $\tau_{c,i}$ is the average effective consumption tax rate.

The results with the labor wedge are in line with those for the labor tax in Table 2. Output volatility is not significant in any of the four regressions. Even the interaction with institutional quality is not significant.

¹²We choose to present the results using the OLS estimates because they typically are more precise than the IV estimates. It is thus more likely to reject our functional form with the OLS estimates.

¹³We also did the exercise of allowing the level relationship between the capital tax and institutional quality to be nonlinear by adding a quadratic main effect (quality)² to the interaction specification. This leaves the quality×volatility interaction essentially unchanged (−0.259, $p < 0.01$ without the tax-revenue control; −0.236, $p < 0.01$ with it), even though the quadratic term is itself significant, so the estimated interaction is not an artifact of unmodeled curvature in the quality main effect.

¹⁴Estimating the same four regressions by two-stage least squares with the instruments of Table 3 gives the same picture: neither volatility nor its interaction with institutional quality is significant in any column.

Table 5: Combined labour–consumption wedge and output volatility: OLS regressions

	Labor wedge $1 - \frac{1 - \tau_\ell}{1 + \tau_c}$			
	(1)	(2)	(3)	(4)
Output volatility	0.007 (0.021)	0.006 (0.015)	0.007 (0.016)	0.019 (0.014)
Institutional quality		0.660*** (0.132)	0.604*** (0.155)	0.237 (0.160)
Quality $\times$ Volatility			−0.182 (0.123)	−0.119 (0.102)
Tax revenue / GDP				0.011*** (0.004)
OECD dummy	0.193*** (0.047)	−0.030 (0.054)	−0.019 (0.062)	0.047 (0.052)
R^2	0.350	0.564	0.578	0.704
Countries	37	37	37	37

Notes. Dependent variable: the combined labor–consumption tax wedge $1 - (1 - \tau_\ell)/(1 + \tau_c)$, where τ_ℓ is the effective labor tax and τ_c the effective consumption tax (built at the country–year level, then averaged). OLS on country means; HC1 robust standard errors in parentheses. Columns (3) and (4): quality and volatility mean-centered. Tax revenue / GDP as in Table 2; the sample is the 37-country labor-tax sample of that table. ***, **, *: 1%, 5%, 10%.

A fourth exercise adds public indebtedness to the list of controls. A government carrying more debt must raise more revenue, and highly indebted economies may also differ in their exposure to aggregate risk, so debt could confound both the level and the interaction effects. Adding general government gross debt as a share of GDP to column (4) of Table 2 leaves the picture intact: debt enters positively and significantly, with a coefficient of 0.001 per percentage point of GDP, the volatility coefficient falls from -0.052 to -0.034 and remains significant at the 5% level, and the interaction is -0.210 and significant at the 1% level. In the two-stage least squares version of column (4) of Table 3, debt absorbs more of the variation the instrument exploits: the volatility coefficient is -0.027 with a p-value of 0.05, and the interaction, at -0.174, is no longer significant, with the conditional first-stage statistic for the interaction falling from 95 to 66. Debt plays no role for the labor tax, where its coefficient is small and insignificant and volatility and the interaction remain insignificant.

Finally, we add to the labor-tax regressions the seven economies for which the labor tax is available but the net capital tax is not. On 44 countries, output volatility remains insignificant in all four specifications of columns (5)–(8) of Table 2, so the third fact does not depend

on the sample. The interaction with institutional quality is somewhat stronger in this larger sample, at -0.179 in column (7) and -0.140 in column (8), both significant at the 5% level, but it is insignificant once debt is added and in every two-stage least squares specification.

Several caveats remain. The instrumental-variables estimates rest on roughly forty country means, so we subject the capital-tax interaction specification estimated by two-stage least squares (column (4) of Table 3) to a battery of small-sample checks. First, weak instruments are not a concern: the Sanderson–Windmeijer conditional first-stage statistics are 102 for the volatility term and 95 for the interaction, and never fall below 46 when any single country is dropped. Second, leave-one-out estimation shows that no single country drives the results: dropping each country in turn keeps the volatility coefficient between -0.064 and -0.042 and the interaction between -0.304 and -0.174, and both remain significant at the 5% level in every case. Third, winsorising both the capital tax and volatility at the 5th and 95th percentiles leaves both coefficients essentially unchanged: the volatility coefficient is -0.049 with a p-value lower than 0.001 and the interaction coefficient is -0.259 and is significant at the 1% level. Finally, a nonparametric case bootstrap (resampling countries, 2,000 replications) confirms the level effect, whose 95% confidence interval $[-0.13, -0.01]$ excludes zero. But, the interaction’s bootstrap interval $[-0.80, 0.07]$ is wider and includes zero, reflecting the efficiency cost of instrumenting both volatility and its interaction in a cross-section of this size.

3 The Model

We now build the model that accounts for the facts in the empirical section. It is an extended version of the neoclassical growth model with political frictions of Acemoglu et al. (2008), in which a self-interested politician sets taxes on labor and capital and diverts public resources for his own consumption, disciplined only by citizens’ ability to replace him. We extend this framework in two directions: aggregate shocks to total factor productivity, and an exogenous parameter capturing the quality of institutions. Their interplay drives our main result: under aggregate risk, it becomes optimal to lower the tax on capital rather than on labor, most so where institutions are strong—precisely the pattern we document in the data.

3.1 Households

A representative household chooses consumption $c_t(s^t)$, labor $n_t(s^t)$, capital $k_{t+1}(s^t)$, and holdings of one-period state-contingent claims $b_{t+1}(s_{t+1} | s^t)$, and, at the end of each period, whether to retain or replace the incumbent politician. We index variables by the history $s^t = (s_0, \dots, s_t)$ of exogenous states realized through date t . The household maximizes the expected lifetime utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t(s^t), n_t(s^t)), \quad (4)$$

taking factor prices and fiscal policy as given, where $\beta \in (0, 1)$ is the discount factor. The period utility u is strictly increasing in consumption, strictly decreasing in labor, strictly concave, and the marginal utility of consumption is convex.

In addition to accumulating physical capital, the household trades a full set of one-period state-contingent claims. It owns the capital stock, which it rents to the competitive firms, and supplies labor to them. The government, described in the detailed in Government subsection, levies a proportional tax $\tau_{\ell,t}(s^t)$ on labor income and $\tau_{k,t}(s^t)$ on capital income net of depreciation. In every history s^t , the household's flow budget constraint is

$$c_t(s^t) + k_{t+1}(s^t) + \sum_{s_{t+1}} p(s_{t+1} | s^t) b_{t+1}(s_{t+1} | s^t) = (1 - \tau_{\ell,t}(s^t)) w_t(s^t) n_t(s^t) + [1 + (1 - \tau_{k,t}(s^t))(r_t(s^t) - \eta)] k_t(s^{t-1}) + b_t(s_t | s^{t-1}) \quad (5)$$

where $w_t(s^t)$ and $r_t(s^t)$ are the wage and the rental rate of capital, respectively; η the depreciation rate, and $p(s_{t+1} | s^t)$ the price of a claim that pays one unit of consumption next period in state s_{t+1} . The household earns the after-tax return $(1 - \tau_{k,t}(s^t))(r_t(s^t) - \eta)$ on it, so the capital-tax base is the return to capital net of depreciation. Initial holdings (k_0, b_0) are given, a no-Ponzi condition rules out unbounded borrowing, and we impose the standard Ramsey restriction $\tau_{k,0} = 0$ that prevents a confiscatory levy on pre-existing capital.

3.2 Firms

In this economy, a representative firm demands labor and capital to households in competitive factor markets. Output $Y_t(s^t)$ is

$$Y_t(s^t) = A_t F(k_t(s^{t-1}), n_t(s^t)), \quad (6)$$

where A_t is the stochastic total factor productivity (TFP) shock—the only source of aggregate uncertainty, whose history is recorded by s^t —and F is strictly increasing and concave in both arguments and exhibits constant returns to scale. The firm maximizes profits taking factor prices as given, so the wage and the rental rate of capital equal the marginal products of labor and capital:

$$w_t(s^t) = A_t \frac{\partial F(k_t(s^{t-1}), n_t(s^t))}{\partial n_t}, \quad (7)$$

$$r_t(s^t) = A_t \frac{\partial F(k_t(s^{t-1}), n_t(s^t))}{\partial k_t}. \quad (8)$$

3.3 Government

In this economy the government is run by a politician who sets taxes and can divert public resources for his own consumption. The politician's expected payoff is

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \delta^t v(g_t(s^t)), \quad (9)$$

where $g_t(s^t)$ is the amount of public resources the politician appropriates, δ the political discount factor, and v is strictly increasing and concave. In addition, an exogenous public expenditure $\bar{G}$ must be financed with taxes, so total public spending in history s^t is

$$G_t(s^t) = \bar{G} + g_t(s^t). \quad (10)$$

As shown in the quantitative section, $\bar{G}$ disciplines the size of the government; it can be interpreted as the cost of a public good that enters neither the household's preferences nor its budget constraint. The government's flow budget constraint is

$$\bar{G} + g_t(s^t) + b_t(s_t | s^{t-1}) = \tau_{k,t}(s^t)(r_t(s^t) - \eta) k_t(s^{t-1}) + \tau_{\ell,t}(s^t) w_t(s^t) n_t(s^t) + \sum_{s_{t+1}} p(s_{t+1} | s^t) b_{t+1}(s_{t+1} | s^t) \quad (11)$$

3.4 Dynamic game

The household and the politician interact in a dynamic game with the following within-period timing:

- The incumbent politician sets the labor and capital tax rates and his appropriation $g_t(s^t)$, anticipating how the household will optimally respond.
- Taking fiscal policy and factor prices as given, the household chooses consumption, labor, investment, and bond holdings; individual households are non-strategic in these market decisions.
- At the end of the period, citizens act jointly—as a single representative citizen—and decide whether to retain the politician or replace him with an identical one.¹⁵

We focus on the best equilibrium from the household’s point of view, in which the politician is never replaced along the equilibrium path, following Acemoglu et al. (2008, 2011) and Yared (2010). The politician cannot commit, so the equilibrium is sustained by the threat of replacement, and his incentives are governed by three values:

- **Punishment value:** a politician who deviates is replaced and obtains $v(0) = 0$ in every subsequent period.
- **Deviation value:** by deviating he seizes a fraction $1 - \theta$ of current output, with payoff $v((1 - \theta) Y_t(s^t))$; the parameter $\theta \in (0, 1]$ indexes the quality of institutions, with higher θ leaving a smaller divertible share.
- **Continuation value:** by complying he obtains $\mathbb{E}_t \sum_{s=0}^{\infty} \delta^s v(g_{t+s})$, the discounted stream of future rents.

The politician therefore complies in every history if and only if

$$\mathbb{E}_t \sum_{s=0}^{\infty} \delta^s v(g_{t+s}) \geq v((1 - \theta) Y_t(s^t)) \quad \forall t \quad (12)$$

We call this the Political Participation Constraint (PPC).

¹⁵Placing the replacement decision at the end of the period follows Acemoglu et al. (2008) and the alternative timing of Yared (2010); it leaves the sustainability logic unchanged.

3.5 Equilibrium

A competitive equilibrium, given a policy $\{\tau_{\ell,t}(s^t), \tau_{k,t}(s^t), g_t(s^t)\}$ and the mandatory spending $\bar{G}$, is a collection of allocations $c_t(s^t)$, $n_t(s^t)$, $k_{t+1}(s^t)$, $b_{t+1}(s_{t+1} | s^t)$ and prices $w_t(s^t)$, $r_t(s^t)$, $p(s_{t+1} | s^t)$ such that, at every date and history s^t :

- the household maximizes its expected utility subject to its budget constraint, taking prices and policy as given;
- firms maximize profits, so that factor prices equal their marginal products;
- the government budget constraint holds;
- markets clear, so that the resource constraint holds.

$$c_t(s^t) + g_t(s^t) + \bar{G} + k_{t+1}(s^t) = A_t F(k_t(s^{t-1}), n_t(s^t)) + (1 - \eta) k_t(s^{t-1}) \quad (13)$$

The primal problem

Politicians enter freely and the incumbent is disciplined by the threat of replacement, so the object of interest is the best sustainable equilibrium: among all competitive equilibria that can be sustained as a subgame-perfect equilibrium of the dynamic game, the one that maximizes the household's welfare subject to the Political Participation Constraint (12).

Our model builds upon two theoretical papers: Acemoglu et al. (2008) and Yared (2010). Acemoglu et al. (2008) characterize the best sustainable equilibrium of a neoclassical growth model with physical capital in which a self-interested politician is disciplined by the threat of replacement, and show that its allocation solves a constrained-planning problem of the kind described above. Their environment, however, is deterministic: fundamentals carry no aggregate uncertainty.

Yared (2010), in turn, characterizes the efficient sustainable equilibrium under aggregate uncertainty. In his economy a stochastic process drives the productivity of public spending, and fiscal policy is contingent on the realized history s^t . There is no physical capital, however, and output is produced from labor alone.

Our setting unites the two: physical capital, as in Acemoglu et al. (2008), together with the aggregate uncertainty that, as in Yared (2010), makes policy state-contingent—here an exogenous stochastic process for total factor productivity. Our claim is that the constrained-planning characterization common to both papers carries over to this combined environment:

adding aggregate risk to the capital economy of Acemoglu et al. (2008), or physical capital to the stochastic economy of Yared (2010), does not alter the characterization of the best sustainable equilibrium. The reason, made precise in the appendix, is intuitive: the aggregate shock enters only through a conditional expectation over histories, which acts linearly and preserves the structure on which both characterizations rest, while a deviating politician's payoff depends only on the current state.

The following lemma states this claim formally.

Lemma 1. *Suppose u is concave and v is strictly increasing. Then, in the presence of physical capital and aggregate productivity risk, the best sustainable equilibrium is well defined, and its allocation can be represented as the solution of a constrained planning problem: the allocation that maximizes the household's welfare among competitive equilibria, subject to the Political Participation Constraint (12).*

The proof is in Appendix C.

Ramsey approach

The lemma allows us to dispense with the fiscal instruments. Consolidating the household's budget constraints with its first-order conditions reduces the requirement that an allocation be a competitive equilibrium to a single intertemporal implementability constraint, derived in Appendix B. Collecting it with the resource constraint and the Political Participation Constraint, the best sustainable equilibrium solves the Ramsey problem

$$(P) \quad \max_{\{c_t, n_t, g_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t(s^t), n_t(s^t)), \quad (14)$$

subject to, at every date t and history s^t , the resource constraint

$$c_t(s^t) + g_t(s^t) + \bar{G} + k_{t+1}(s^t) = A_t F(k_t(s^{t-1}), n_t(s^t)) + (1 - \eta) k_t(s^{t-1}),$$

the implementability constraint

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u_c(c_t(s^t), n_t(s^t)) c_t(s^t) + u_n(c_t(s^t), n_t(s^t)) n_t(s^t)] = u_c(c_0, n_0) \{ [1 + (1 - \tau_{k,0})(r_0 - \eta)] k_0 + b_0 \} \quad (15)$$

and the Political Participation Constraint

$$\mathbb{E}_t \sum_{s=0}^{\infty} \delta^s v(g_{t+s}) \geq v((1-\theta) Y_t(s^t)).$$

Under the Ramsey restriction $\tau_{k,0} = 0$ the initial-wealth term in the implementability constraint reduces to $u_c(c_0, n_0) [(1-\eta+r_0)k_0 + b_0]$.

3.6 Recursive formulation

Problem (14) is not a standard dynamic program. The Political Participation Constraint (12) is forward looking: the value of complying today is the discounted expected politician's utility of all future rents, so today's choice of g carries an implicit promise about future consumption and the problem does not satisfy a Bellman equation in its standard version. To formulate such problem in its recursive form, we follow Marcet and Marimon (2019). Their method dualizes the forward-looking constraint in a saddle-point Lagrangian and introduces a co-state—a running record of the multipliers on that constraint—so that the planner's value satisfies a saddle-point functional equation, the analog of the Bellman equation. Here the augmented state is (k, γ, A) : the physical capital k , the productivity state A , and the co-state γ that summarizes the history of the PPC. The algebraic transformation from (14) to the recursive form below is in Appendix D.

Let ψ be the multiplier on the implementability constraint (15), set once at $t = 0$ and held fixed thereafter, and let μ be the multiplier on the PPC in the current period. The co-state γ enters as a state variable and is updated each period by the current PPC multiplier. Writing primes for next-period objects, the best sustainable equilibrium solves the saddle-point functional equation

$$V(k, \gamma, A) = \min_{\mu \geq 0} \max_{c, n, g, k'} \left\{ u(c, n) + \gamma' v(g) - \mu v((1-\theta) A F(k, n)) - \psi [u_c(c, n) c + u_n(c, n) n] + \beta \mathbb{E}[V(k', \gamma', A')] \right\} \quad (16)$$

subject to the resource constraint

$$c + g + \bar{G} + k' = A F(k, n) + (1-\eta) k, \quad (17)$$

and the co-state law of motion

$$\gamma' = \frac{\delta}{\beta} \gamma + \mu. \quad (18)$$

The minimization over μ and maximization over (c, n, g, k') define the saddle point; γ is

initialized at $\gamma_0 = 0$, so that the co-state simply accumulates the discounted history of binding PPCs. Intuitively, γ' records how much weight the politician's past and present participation constraints place on current rents: when the PPC binds ($\mu > 0$) the weight γ' on $v(g)$ rises, and it decays at the rate δ/β that reflects the gap between the political and household discount factors. This turns the history-dependent promise embedded in (12) into a Markov state.

Let $\Gamma \equiv u_{cc}(c, n)c + u_c(c, n)$ and $\Lambda \equiv u_{nn}(c, n)n + u_n(c, n)$ collect the derivatives of the implementability integrand with respect to c and n . The first-order conditions of the saddle point (16) are, for consumption,

$$u_c(c, n) - \psi \Gamma = \gamma' v_g(g), \quad (19)$$

for labor,

$$u_n(c, n) + [u_c(c, n) - \psi \Gamma] A F_n(k, n) - \psi \Lambda = \mu v_g((1 - \theta) A F(k, n)) (1 - \theta) A F_n(k, n), \quad (20)$$

the intertemporal Euler equation,

$$u_c(c, n) - \psi \Gamma = \beta \mathbb{E} \left[(u_c(c', n') - \psi \Gamma') (1 - \eta + A' F_k(k', n')) - \mu' (1 - \theta) v_g((1 - \theta) A' F(k', n')) A' F_k(k', n') \right] \quad (21)$$

and the PPC, which in recursive form reads

$$v(g) + \delta \mathbb{E} [v((1 - \theta) A' F(k', n'))] \geq v((1 - \theta) A F(k, n)), \quad (22)$$

holding with complementary slackness against $\mu \geq 0$, where Γ' and $F_k(k', n')$, $F_n(k', n')$ denote the next-period values of Γ and the marginal products.

We read these conditions as follows. Equation (19) is the intratemporal optimality condition linking the marginal value of private consumption, net of its implementability cost $\psi \Gamma$, to the marginal value of the politician's consumption $\gamma' v_g(g)$; the co-state γ' acts as the relative price between private and political consumption, so that the more weight the history of binding PPCs has placed on rents, the more is diverted to g .

Equation (20) is the labor-supply margin. When the PPC does not bind ($\mu = 0$) and the implementability constraint is absent ($\psi = 0$), it collapses to $u_n + u_c A F_n = 0$, the standard marginal-rate-of-substitution condition of the undistorted neoclassical model. The distortions on this margin therefore arise solely from two sources: the implementability terms in ψ , which capture the Ramsey cost of raising revenue, and the term in μ , which is

the politician's temptation to divert. The latter is scaled by $(1 - \theta)$: the larger the divertible share $1 - \theta$, the more output the politician would seize by deviating, and the more the labor margin is bent to relax the PPC. As institutions strengthen, $\theta \rightarrow 1$, the divertible share vanishes and this wedge disappears.

Equation (21) is the Euler equation. When the political constraint is slack everywhere ($\mu' = 0$), it reduces to the undistorted intertemporal condition in which the net-of-depreciation return $A'F_k(k', n') - \eta$ prices the marginal unit of capital. The final term, in μ' , is the wedge the politician introduces on capital accumulation. It enters the return through the net base $A'F_k(k', n')$ and is again scaled by the divertible share $(1 - \theta)$, so that it vanishes as institution quality improves, $\theta \rightarrow 1$, where nothing can be diverted and the deviation payoff $v((1 - \theta)A'F(k', n')) \rightarrow v(0) = 0$. In the nonstochastic case, where $\gamma' = \gamma$ in the long run, the wedge also vanishes as $\delta \rightarrow \beta$, recovering the Chamley–Judd result of Acemoglu et al. (2011): when the politician is as patient as the household, the intertemporal margin is left undistorted.

Equation (22) is the recursive PPC. It states that the value of complying—current rents $v(g)$ plus the discounted continuation value of the politician's program—must weakly exceed the value of deviating today, $v((1 - \theta)A'F(k, n))$. Written this way, the value of deviating equals the value of complying now and deviating one period later, an equivalence that must hold at every date and state, so that along the equilibrium path the politician never finds it optimal to deviate.

3.7 Taxes

The recursive problem delivers an allocation but no explicit fiscal instruments. We now recover the taxes that decentralize this allocation. We obtain both taxes from the household's optimality conditions evaluated at the planner's allocation¹⁶. Throughout, u_c , u_n , F_k and F_n are evaluated at the recursive solution, $r = AF_k$ is the gross rental rate, $w = AF_n$ the wage, and primes denote next-period objects, with z the current Markov state, z' its successor, and $\pi(z' | z)$ the transition probability.

¹⁶Taxes are not recovered from any government budget identity: the period-by-period government budget might be slacked, as it is implied, once the implementability constraint holds, so it carries no additional information about the tax rates.

Labor tax

The labor tax follows directly from the intratemporal optimality condition. At the competitive equilibrium the household equates its marginal rate of substitution between consumption and labor to the after-tax wage, $-u_n/u_c = (1 - \tau_\ell)w$, with $w = AF_n$. Solving for the tax rate,

$$\tau_\ell = 1 + \frac{u_n}{u_c AF_n}. \quad (23)$$

The labor tax is a static intratemporal wedge: it depends only on contemporaneous quantities, through the plain marginal utility u_c in the denominator, and inherits no intertemporal or precautionary term. It is therefore insensitive to aggregate volatility, which is the model counterpart of the empirical finding that volatility leaves the labor tax unaffected.

Capital tax

The capital tax is recovered by matching the household's intertemporal optimality condition to the planner's. The household's consumption Euler equation, with a tax on the return to capital net of depreciation, is

$$u_c = \beta \mathbb{E}[u'_c((1 - \tau'_k)(r' - \eta) + 1) \mid z], \quad (24)$$

where the taxed base is the net return $r' - \eta$. The recursive (Marcet–Marimon) planner satisfies an analogous Euler equation that, once written in the same marginal-utility units, reads:

$$u_c = \beta \mathbb{E}[u'_c(1 - \eta + r') - \Xi' \mid z]. \quad (25)$$

The term Ξ' is the capital-tax wedge. It collects the contributions of the resource constraint, the implementability constraint and the Political Participation Constraint, weighted by their multipliers, and is therefore a known function of the recursive solution—the co-state γ' and the multipliers μ' and ψ on the participation and implementability constraints, the institutional parameter θ , and the marginal products entering r' .¹⁷ Subtracting the decentralized

¹⁷Explicitly, $\Xi' = \mu'(1 - \theta)v_g((1 - \theta)A'F(k', n'))r' + \psi[\Gamma'(1 - \eta + r') - \Gamma/\beta]$, with $\Gamma \equiv u_{cc}c + u_c$ and Γ' its next-period analogue. The first term is the political wedge: it scales with the multiplier μ' on the Political Participation Constraint and with the divertible share $1 - \theta$, and it vanishes as $\theta \rightarrow 1$ —the strongest-institutions limit in which nothing can be diverted—and, in the nonstochastic economy, as $\delta \rightarrow \beta$, recovering the zero long-run capital tax of Acemoglu et al. (2011). The second term is the implementability wedge; at the deterministic steady state the household's Euler equation gives $(1 - \tau_k^{SS})(r^{SS} - \eta) = 1/\beta - 1$, so the bracket equals $\tau_k^{SS}(r^{SS} - \eta)$ and the implementability term contributes a constant share $\psi(1 - \sigma_c)$ of the steady-state tax—about one tenth at our calibration—so the level is predominantly political (Lemma 2). The derivation of Ξ' is in Appendix D.

Euler (24) from the planner's Euler (25) leaves the matching condition

$$\mathbb{E}[u'_c \tau'_k (r' - \eta) | z] = \mathbb{E}[\Xi' | z], \quad (26)$$

which, written over the Markov states, is the state-contingent condition

$$\sum_{z'} \pi(z' | z) u_c(z') \tau_k(z') (r(z') - \eta) = \sum_{z'} \pi(z' | z) \Xi(z'). \quad (27)$$

The tax falls on the net return $r(z') - \eta$ throughout, consistent with the household budget constraint.

Ex-ante versus ex-post capital tax

Condition (26) ties down only a conditional expectation of the capital tax, not its realization in each successor state. This is the indeterminacy of Chari et al. (1994): the household's Euler equation pins down the ex-ante capital wedge but not the ex-post, state-by-state capital tax. The wedge can be split in many ways between state-contingent capital taxes and state-contingent returns on debt, so any vector $\{\tau_k(z')\}$ that satisfies (27) is one admissible decentralization among an equivalence class; a state-contingent capital tax substitutes for, rather than precludes, state-contingent debt. This indeterminacy issue is discussed in details by (Chari et al., 1994).

We therefore, following (Chari et al., 1994), take as our preferred object the ex-ante capital tax, the conditional mean wedge

$$\bar{\tau}_k(z) = \frac{\mathbb{E}[\Xi' | z]}{\mathbb{E}[u'_c (r' - \eta) | z]}, \quad (28)$$

which is invariant to the debt/capital-tax decentralization and is the level on which the model's volatility and institutional comparisons are based. We also report the ex-post, state-contingent tax $\tau_k(z')$ as one convention-selected member of the indeterminacy class, recovered by solving the linear system (27) for the vector $\{\tau_k(z')\}$ across successor states z' .

4 Comparative statics

This section establishes what the model delivers analytically and what it provably cannot. Without aggregate risk the economy inherits the characterization of the deterministic

political-economy literature: the Political Participation Constraint binds, and it sustains a capital tax that is predominantly political and vanishes as institutions become perfect. Under aggregate risk that characterization fails in a precise sense, and what replaces it is a statement about pricing: risk enters the capital tax through a covariance whose sign is unambiguous, and it enters only through the political part of the tax. How large the resulting reduction is, and how the two forces resolve, cannot be settled by these results; that is the task of the quantitative section.

We begin with the benchmark without risk. In the deterministic version of our economy the logic of Acemoglu et al. (2008, 2011) applies: the threat of replacement disciplines the politician, the constraint binds in the long run, and the sustained capital tax reflects the political friction, disappearing when the politician is as patient as the household. Our environment adds the implementability constraint of distortionary finance, so the exact statement of the benchmark takes the following form; the binding argument follows the papers above, while the tax characterization is specific to our economy.

Lemma 2 (the steady-state capital tax). *Consider the economy without aggregate risk and let $\theta < 1$. At any steady state with finite rents: (i) the Political Participation Constraint binds, with multiplier $\mu^{SS} = (1 - \delta/\beta)\gamma^{SS} > 0$; (ii) the capital tax splits into constant shares of its two components: $\bar{\tau}_k^{imp,SS} = \psi(1 - \sigma_c)\tau_k^{SS}$, so the political share is $1 + \psi(\sigma_c - 1)$; (iii) $\tau_k^{SS} \rightarrow 0$ as $\theta \rightarrow 1$, and $\tau_k^{SS} \rightarrow 0$ as $\delta \rightarrow \beta$.*

The lemma pins down the anchor that risk will erode. Part (i) says the friction is active: a slack constraint would give the politician no rents, making the one-shot deviation strictly profitable, so sustaining the equilibrium requires rents and a binding constraint. Part (ii) quantifies the composition of the tax: with $\psi < 0$ pinned by $\bar{G} > 0$, the implementability share equals $|\psi|(\sigma_c - 1)$ — about one tenth at our calibration — and vanishes as government spending goes to zero, so the steady-state tax is predominantly political and purely political in the small-government limit. Part (iii) gives the two limits in which the tax disappears: perfect institutions, where nothing can be diverted, and a patient politician, which is the Chamley–Judd limit of Acemoglu et al. (2011). Between the limits the level declines monotonically with institutional quality in the computed model.

The deterministic characterization rests on a fixed point of the co-state, and aggregate risk removes it. The next result makes this precise.

Proposition 1 (aggregate risk breaks the deterministic characterization). *In the economy with aggregate risk: (i) along any solution, the co-state is constant at γ^* if and only if the multiplier is constant at $\mu^* = (1 - \delta/\beta)\gamma^*$; (ii) on any date at which the constraint is slack, the co-state strictly decays: $\gamma_{t+1} = (\delta/\beta)\gamma_t < \gamma_t$ whenever $\gamma_t > 0$; (iii) under any stationary*

ergodic solution, $\mathbb{E}[\mu] = (1 - \delta/\beta) \mathbb{E}[\gamma]$, strictly positive whenever the constraint binds with positive probability.

The proposition is how our extension relates to the deterministic literature. By part (i), a constant co-state requires a multiplier that does not move with the state; in the computed model the multiplier varies across states at every parameter combination we solve, so the steady-state object on which the deterministic characterization rests does not exist under risk. By part (iii), what survives is the average: the friction remains active in the mean. The deterministic results are therefore not wrong but incomplete under aggregate risk — they describe the average of an economy whose state-by-state behavior is genuinely different, and it is in that state-by-state behavior that the volatility effects of this paper live.

What replaces the fixed point is a pricing result. The capital tax is the expected wedge divided by the expected marginal-utility-weighted net return, and risk affects this ratio in a way whose direction can be signed.

Proposition 2 (the decomposition and the pricing of the political wedge). *At every state z : (i) the capital tax decomposes exactly as $\bar{\tau}_k(z) = \bar{\tau}_k^{pol}(z) + \bar{\tau}_k^{imp}(z)$, the two components sharing the denominator $\mathbb{E}[u'_c(r' - \eta) \mid z]$, where the numerators are the conditional expectations of the political and implementability terms of the wedge Ξ' ; (ii) holding the allocation fixed, θ enters the wedge only through the political term; (iii) suppose that, at the state z , consumption and output move together across next period's productivity states at the chosen savings: $(c'_s - c'_t)(y'_s - y'_t) \geq 0$ for every pair of successor states. Then $\text{Cov}(\Gamma', r' \mid z) \geq 0$ and hence $\psi \text{Cov}(\Gamma', r' \mid z) \leq 0$: the covariance term that risk introduces into the pricing of the tax is weakly negative at every state, and zero without risk.*

Part (iii) is the theoretical core of the volatility effect. The covariance exists only under risk, its sign is unambiguous under the stated comonotonicity property — a monotonicity condition on the computed policies, which holds in the solved model and is the only non-primitive ingredient of the result — and it pushes the capital tax down. The economics is the hedging property documented in the quantitative section: capital pays off in the states in which marginal utility is low, so risk lowers the value of the claim that the tax prices. Note what part (iii) does not claim: it signs one term of the pricing, not the total effect of volatility on the tax.

Parts (ii) and (iii) together deliver the structural version of the interaction between risk and institutions.

Corollary 1 (the interaction is political). *Institutions and risk interact only through the political component of the capital tax: the part of the tax that institutional quality disciplines*

is the same part that risk reprices.

The corollary is the model’s qualitative counterpart of the second fact of the empirical section. Institutions determine how much political tax exists; risk determines how much of it survives the pricing; and neither force operates through the implementability component, which carries no θ and no risk-signed term. Whether the resulting proportional reduction rises or falls with institutional quality, and by how much, depends on how strongly risk erodes the political component at each level of the political tax — an equilibrium magnitude.

That magnitude is exactly what these results cannot deliver, and the limitation is provable rather than an artifact of our methods. The sign of the total volatility effect combines the covariance of Proposition 2 with terms of the same order whose signs depend on the equilibrium allocation, so no ranking of the two forces follows from the primitives; and the size of the reduction depends on the ergodic distribution of the economy, which no comparative static reaches. The quantitative section takes over at this point: it measures the covariance, the components, and the reduction at a disciplined calibration, and finds that the political component carries the entire response.

Proofs are in Appendix E.

5 Quantitative Results

This section disciplines the model with data from two reference economies and reports what the mechanism delivers in each. We calibrate the model separately to the United States, which represents the advanced world, and to Chile, which represents the emerging one, fixing the political discount factor from leader-tenure data and choosing the remaining parameters so that each economy matches its capital-output ratio, the volatility of its output, its average effective capital tax, and the size of its government. In the United States aggregate risk holds the capital tax 21 percentage points below its no-risk level, the reduction deepens as output volatility rises, the reduction is largest where institutions are strong, and the labor tax does not move. In Chile the capital tax never moves by more than 0.81 percentage points across the same sweeps, which is what the regressions of the empirical section find for countries with weak institutions, and decomposing the tax shows why the two economies differ: the implementability wedge of the standard Ramsey problem responds to neither risk nor institutions, so everything the capital tax does operates through the Political Participation Constraint.

5.1 Two reference economies

We calibrate the model twice, once for an advanced economy and once for an emerging one. The United States is the natural advanced benchmark and the economy for which the model’s data requirements are best met. Chile is our emerging reference for three reasons. First, its output is more volatile than the United States’, 1.8 against 1.3 percent, and its effective capital tax is a third of the United States’, so the two economies sit at opposite ends of the relation that Section 2 documents. Second, it has national accounts detailed enough to construct the same effective tax rates as for the United States, and Penn World Table series that discipline the capital-output ratio and the capital share on the same footing. Third, its leader tenure is recorded in the same database, so every object that differs across the two calibrations is measured the same way in both. Chile is a conservative choice among emerging economies, since its volatility sits near the middle of our sample rather than at the top; the lessons of the exercise are therefore about the emerging world rather than about the extremes of it.

The political discount factor is fixed from data on political turnover in each economy. We set δ equal to the annual probability that the chief executive remains in office, computed from the Archigos database of political leaders (Goemans et al., 2009): 0.821 for the United States and 0.837 for Chile. The politician therefore discounts the future by his probability of surviving in office, which ties his horizon to an observable political object and reserves the tax moments for the institutional parameter θ .¹⁸

The utility and technology block is held fixed and, where the data measure it by country, set country by country. We set $\sigma_c = 2$, depreciation $\eta = 0.10$, persistence $\rho = 0.9$ of the productivity process, and $\nu = 0.7$ in the politician’s utility. The capital share is the one the Penn World Table measures for each economy, 0.36 for the United States and 0.545 for Chile.¹⁹

The remaining four parameters are chosen, in each economy, so that four model moments match their data counterparts. The calibrated block is $(\beta, \sigma_A, \theta, \bar{G})$, where σ_A is the standard deviation of productivity innovations. The targets are the capital-output ratio, the volatility

¹⁸Fixing δ externally is also what identifies θ : the model’s moments pin down δ and θ only through a single index, in the sense that a continuum of (δ, θ) pairs reproduces the same moments to four decimal places, so a joint fit would report an arbitrary point on that ridge.

¹⁹We set $\nu = 0.7$ for both economies because lower values cannot accommodate Chile. At $\nu = 0.5$ and $\nu = 0.6$ the Political Participation Constraint does not generate the political wedge that Chile’s capital tax requires at its observed leader tenure, even with institutions at their weakest, so the fit runs into the boundary of the parameter space; $\nu = 0.7$ is the lowest value at which both economies are matched with an interior institutional quality.

of output, the mean effective capital tax of the empirical section, and the government-consumption share used as a control in the regressions of the empirical section.

Two measurement conventions align the model’s moments with the data’s. Model output is filtered exactly as the data is before computing its volatility, so the volatility target compares like with like.²⁰ And because policy is evaluated on simulated paths, every moment is computed on the model’s ergodic distribution, with the capital tax measured by the median of the ex-ante tax of the Taxes subsection.²¹

Table 6 shows that the fit is tight on all four targeted moments in both economies. Output volatility is matched to within 0.01 percentage points, the government share to within 0.2 points, and the capital-output ratio to within 0.04. The capital tax is matched exactly in Chile and to within 1.3 percentage points in the United States, where the calibrated β sits at the upper end of its admissible range: reproducing an average capital tax of 0.35 under a tenure-disciplined political horizon requires patience close to the golden rule.

The calibrated institutional quality is higher in the United States than in Chile, 0.81 against 0.31, and it is worth being explicit about what that number is and is not. In the model θ is the share of output that a deviating politician cannot seize, so it is a statement about the rents that the political equilibrium concedes, and it is identified here from the capital tax that each economy collects given its other fundamentals. It is not the institutional quality index of Section 2, which is built from governance and rule-of-law indicators, and the two need not agree country by country. We use the index to order countries in the empirical section and θ to discipline the politician in the model, and we compare the two only through the response of the capital tax to risk, which is the object both sections measure.

The one notable miss is on an untargeted moment, the labor wedge, and it goes in opposite directions in the two economies. The model delivers 0.234 for the United States against 0.305 in the data, and 0.229 for Chile against 0.199. The United States gap is not an artifact of omitted government debt: adding initial debt to the implementability condition raises the model’s labor wedge, but matching the data would require initial debt of roughly thirty-six times GDP, so we report the undershoot as a limitation of the labor block rather than repair it.

²⁰The filter and smoothing parameter are those of the empirical section (Ravn and Uhlig, 2002).

²¹Medians rather than means: near the calibrated capital stock the expected net return is close to zero, so ergodic means of tax ratios are dominated by a few extreme draws while medians are stable. Reported medians pool five simulated histories of 18,000 periods each, discarding the first 2,000 periods of each.

Table 6: Calibration to two reference economies

	United States		Chile	
<i>Fixed parameters</i>				
Risk aversion σ_c	2		2	
Depreciation η	0.10		0.10	
Persistence ρ	0.90		0.90	
Politician's curvature ν	0.70		0.70	
Capital share α	0.360		0.545	
Leader retention δ	0.821		0.837	
<i>Calibrated parameters</i>				
Discount factor β	0.9988		0.9586	
Productivity innovation σ_A	0.0203		0.0268	
Institutional quality θ	0.81		0.31	
Government spending $\bar{G}$	0.111		0.083	
<i>Targeted moments</i>				
	Model	Data	Model	Data
Capital-output ratio	3.51	3.54	3.66	3.65
Output volatility	0.0132	0.0132	0.0177	0.0178
Capital tax $\bar{\tau}_k$	0.363	0.350	0.122	0.122
Government share	0.155	0.155	0.126	0.126
<i>Untargeted</i>				
Labor wedge	0.234	0.305	0.229	0.199
No-risk capital tax	0.575	–	0.122	–

Notes. θ is reported in the convention of the model section: the politician diverts a fraction $1 - \theta$ of output, so higher θ means better institutions. Data targets: capital-output ratio and capital share from the Penn World Table 10.01 (Feenstra et al., 2015), at constant national prices over 1995–2019; output volatility and the effective tax rates as constructed in the empirical section; government share is the WDI general-government consumption control of the regressions. The capital tax moment is the ergodic median of the ex-ante tax; the no-risk steady state is reported for reference. The labor wedge combines the labor and consumption taxes of the empirical section, $1 - (1 - \tau_\ell)/(1 + \tau_c)$, because the model has no consumption tax and its labor tax is the whole wedge on the labor margin. Chile's effective capital tax is constructed for the years its national accounts report the required detail.

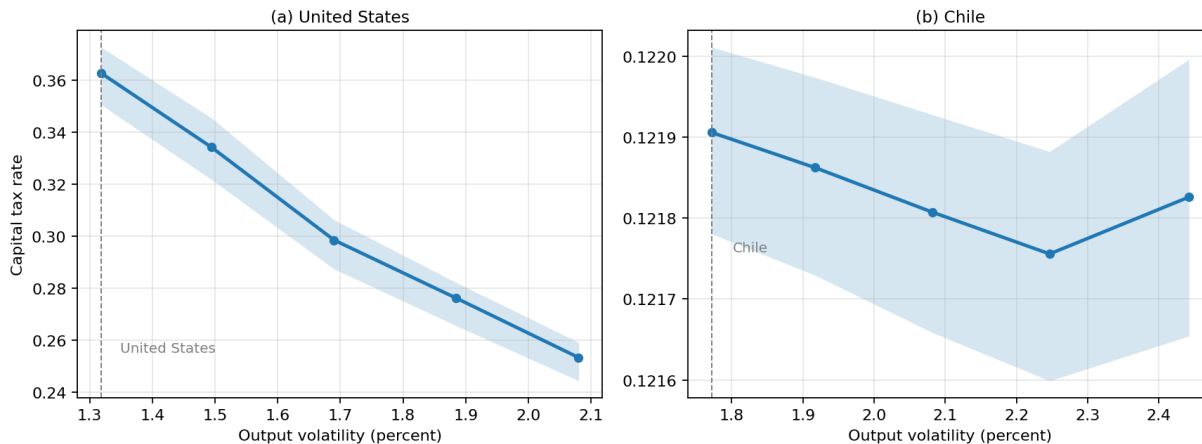


Figure 4: The capital tax and output volatility in the model. The capital tax on the model’s simulated path as the volatility of productivity rises from its calibrated value, for the United States (left panel) and Chile (right panel); the horizontal axis is the resulting volatility of output, rising through the cross-country distribution. Bands show the variation across simulated histories. The vertical scales differ between panels.

5.2 Aggregate risk lowers the capital tax

At the United States calibration, aggregate risk removes 21 percentage points of the capital tax that the political friction would otherwise generate. With productivity risk shut down, the model’s steady-state capital tax is 0.575; at the calibrated volatility, the capital tax on the model’s simulated path is 0.363. Risk of the magnitude observed in United States output is therefore not a small perturbation to the political equilibrium, and this reduction is the model’s counterpart of the first fact of the empirical section.

Moving output volatility up through the cross-country distribution deepens the reduction along a smooth schedule. The left panel of Figure 4 raises the volatility of productivity innovations so that the model’s output volatility spans 1.3 percent, the United States value and the twenty-sixth percentile of the sample, to 2.1 percent, which exceeds the volatility of three fifths of the countries in it. The capital tax falls from 0.363 to 0.253 along this range, so at the upper end risk has removed 32 percentage points of the no-risk tax.²²

The size of the effect is comparable to what the regressions measure. The implied slope of the capital tax with respect to output volatility is -0.14 per percentage point, the same

²²We report the model over the range of volatilities in which the policy iteration converges to an allocation whose simulated path stays in the interior of the capital grid. Beyond that range, which reaches 2.1 percent output volatility at the United States calibration and 2.4 percent at the Chilean one, the iteration settles on a low-capital allocation whose ergodic distribution never reaches the deterministic steady state, and we do not report those solutions.

order of magnitude as the regression coefficient of -0.06 in column (1) of Table 2. We read this as the right quantitative scale rather than a precise match: the model slope is identified from a single country’s calibrated economy, while the regression averages across economies whose institutions differ.

5.3 Institutions govern the strength of the effect

The reduction of the capital tax rises with the quality of institutions over most of its range, which is the model’s counterpart of the second fact. The left panel of Figure 5 solves the model on a grid of institutional quality θ , holding every other parameter at its United States value, and plots the capital tax at each volatility level that solves across the institutional range, together with its value in the absence of aggregate risk.

Two properties of the figure carry the result. First, the level of the capital tax falls as institutions improve at every volatility, inheriting the shape of the no-risk political level. Second, the vertical distance to the no-risk line, which is the reduction that risk produces, is a growing share of that level as θ rises: at United States volatility there is no reduction at all at $\theta = 0.05$, the reduction is 37 percent of the no-risk tax at the calibrated $\theta = 0.81$, and 44 percent at $\theta = 0.86$. At the weakest institutions the two lines meet: risk leaves the capital tax where the political friction puts it, and the economy that sustains the largest political tax is the one whose tax responds least to risk.

The empirical interaction is estimated in levels, and bridging the two objects requires one sentence of care. In the data, the slope of the capital tax with respect to volatility steepens with institutional quality. In the model, the reduction is largest relative to the political level that institutions leave in place, while the level response itself is largest where institutions are weak, simply because weak institutions sustain a much larger tax.

The reason we compare on the proportional margin is that the level belongs to forces outside the model. The level of the capital tax across countries is shaped by fiscal capacity, enforcement, and the tax-base measurement discussed in the empirical section, which is why the identified empirical object is the response of the tax to volatility and its interaction with institutions rather than the level. On the response margin the model and the data agree: risk erodes capital taxation most, in proportional terms, exactly where institutions are strong.

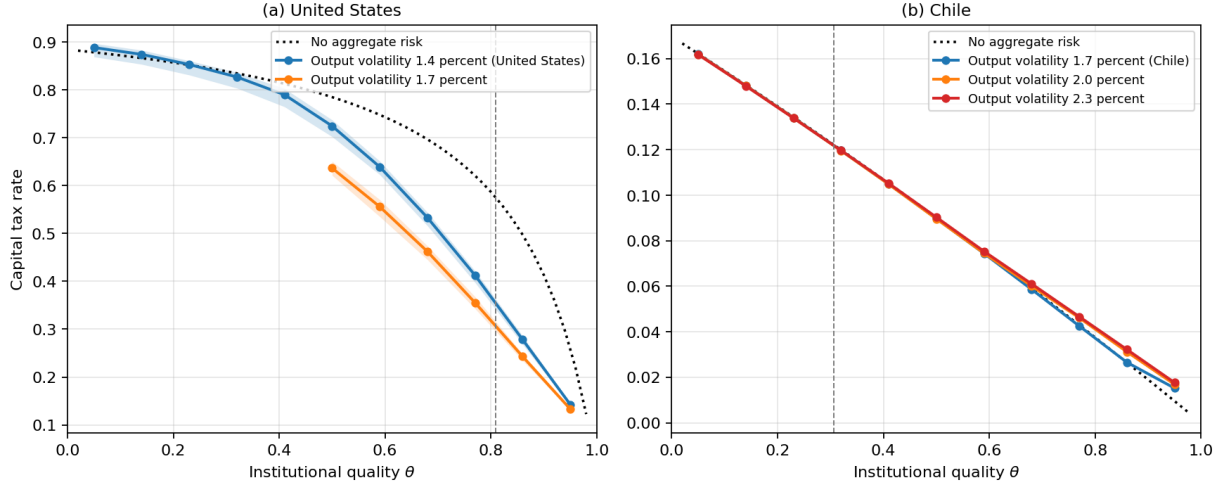


Figure 5: The capital tax across institutional quality and volatility. Each solid line shows the capital tax on the model’s simulated path at a given level of output volatility, across institutional quality, for the United States (left panel) and Chile (right panel); the dotted line is the tax in the absence of aggregate risk. Bands show the variation across simulated histories; the dashed vertical line marks each economy’s calibrated θ . The vertical scales differ between panels.

5.4 Capital, not labor

Everywhere on the grid, risk moves the capital tax and leaves the labor tax where it stands, which is the model’s counterpart of the third fact. Across all the combinations of institutional quality and volatility that we solve at the United States calibration, the labor wedge never moves by more than 0.1 percentage points when volatility rises, against a fall of 21 to 32 points in the capital tax over the same range (Figure 6). The same holds in the Chilean calibration, where the labor wedge moves by 0.1 points.

The asymmetry is the quantitative expression of the tax formulas of the model section. The labor tax is a static wedge built from contemporaneous marginal utilities, with no intertemporal term through which the distribution of future states could enter, while the capital tax prices next period’s wedge and is therefore exposed to risk. In the data, output volatility carries a significant coefficient only for the capital tax; in the model, volatility has a margin to operate on only for the capital tax.

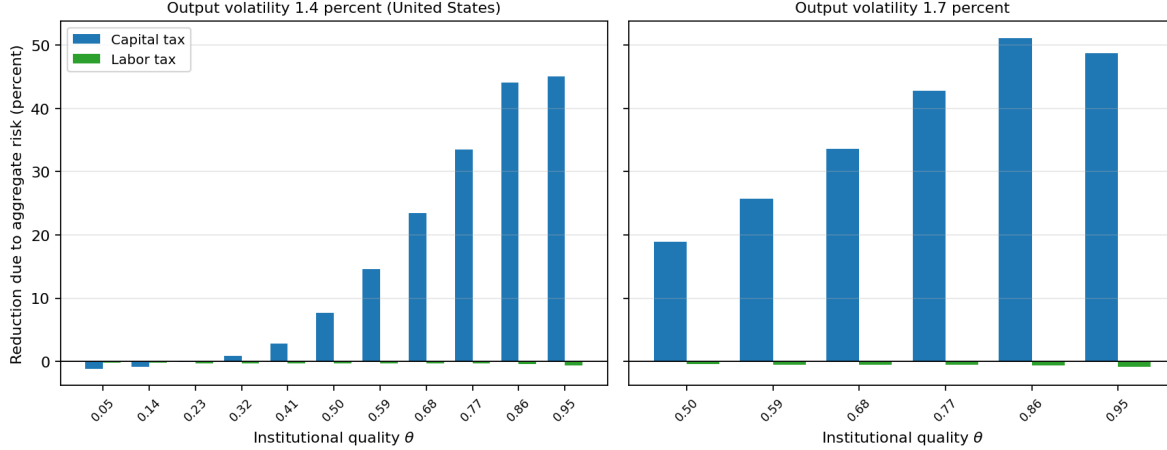


Figure 6: The reduction in taxes caused by aggregate risk, capital versus labor. Bars show how much lower each tax is on the model’s simulated path than in the absence of aggregate risk, as a share of its no-risk value, across institutional quality, at the United States calibration: at United States output volatility (left) and at the top of the empirical volatility range (right).

5.5 The emerging economy

In the Chilean calibration the capital tax barely responds to risk, at any level of volatility or institutional quality we solve. Without risk Chile sustains a capital tax of 0.122; at its calibrated volatility the tax on the simulated path is 0.122, and raising output volatility to 2.4 percent, which exceeds the volatility of six sevenths of the sample, lowers it to 0.122. The right panel of Figure 4 shows the schedule, on a vertical scale far finer than the left panel’s. Across the institutional grid of the right panel of Figure 5 the picture is the same: institutions move the level of the tax, from 0.027 at $\theta = 0.86$ to 0.162 at $\theta = 0.05$, and risk moves it by 0.81 percentage points at most.

The precautionary motive is present in Chile, and it operates on quantities rather than on the tax. Under risk the Chilean economy holds 0.3 percent more capital than it would without risk, a buffer stock as large as the 0.3 percent the United States economy accumulates, and its household’s consumption is as smooth relative to output as the United States household’s. What differs is the price at which that buffer is held. In Chile capital earns 4.9 percent a year net of depreciation, so a bad productivity draw leaves capital profitable and the tax that supports the politician’s wedge has no reason to move; in the United States the same net return is 0.3 percent, so bad draws push the return on capital below zero and the tax falls into subsidy in 23 percent of periods.

This pattern is the one the regressions of the empirical section report for countries with

weak institutions. There the estimated response of the capital tax to volatility is small and statistically indistinguishable from zero, while it is large and precisely estimated where institutions are strong. The model reproduces both halves at calibrations disciplined by each economy’s own data, and it locates the difference in the return that capital earns rather than in the strength of the precautionary motive itself.

5.6 The mechanism: the political wedge carries the risk response

The capital tax splits exactly into a political and an implementability component, and only the political component responds to anything this paper varies. The recursive formulation prices the capital tax from the wedge Ξ' , which is the sum of a political term, proportional to the multiplier on the Political Participation Constraint, and an implementability term, the standard cost of financing government spending with distortionary taxes. Dividing each term by the common expected-return factor decomposes the tax, state by state and exactly, into $\bar{\tau}_k = \bar{\tau}_k^{pol} + \bar{\tau}_k^{imp}$.

Figure 7 reports the two components across institutional quality, and the division of labor between them is complete. The implementability component stays inside the band $[0.07, 0.09]$ at every institutional level and every volatility we compute: it is flat in θ , because institutions enter the wedge only through the political term, and it is flat in risk. The political component is everything else. It generates the level of the tax and its gradient in institutions, and its fall accounts for the entire reduction of the capital tax as volatility rises, at every point of the grid.

The political component does not fall because the constraint loosens. On the ergodic distribution the multiplier on the Political Participation Constraint is flat in volatility and the constraint binds in every simulated period. The wedge’s own ingredients, from the multiplier to the politician’s marginal value of diversion to output and capital, are likewise unmoved by risk.

What moves is the valuation term that converts the wedge into a tax. The conditional covariance between the household’s marginal utility and the return to capital strengthens with volatility, by the same amount at every institutional level, while the expected net return at the calibrated capital stock is close to zero. Capital pays off in good states, when marginal utility is low, so risk makes capital a poor hedge, and the tax that supports the politician’s wedge is driven down; in a growing share of states it is driven below zero, so that high volatility polarizes the capital tax into taxation and subsidy regimes. This is the

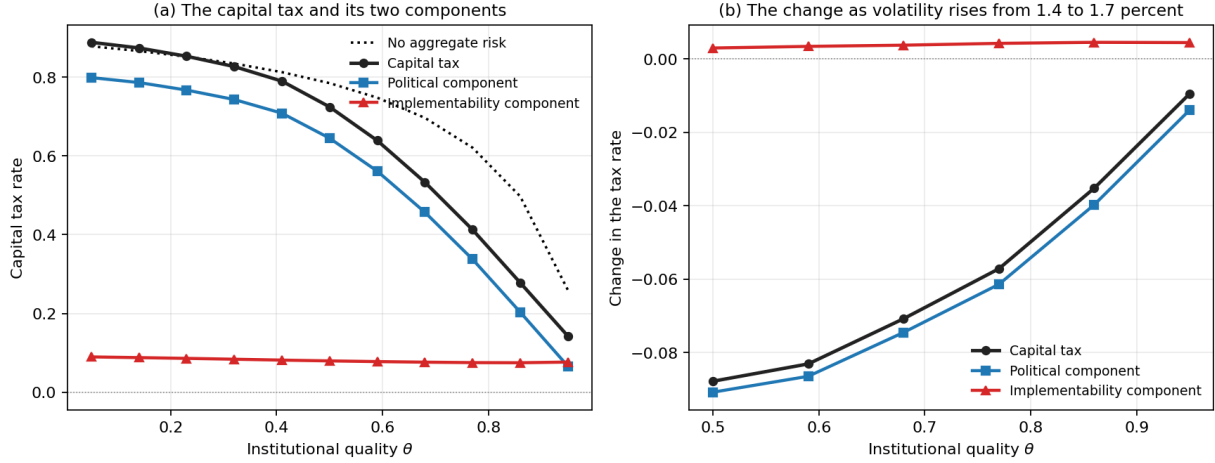


Figure 7: Decomposing the capital tax. Panel (a) splits the capital tax at United States volatility into its political and implementability components, across institutional quality; the dotted line is the tax in the absence of aggregate risk, which is predominantly political (Lemma 2). Panel (b) shows how each component changes when output volatility rises from 1.4 to 1.7 percent, over the institutional range in which both solve: the political component accounts for the entire response.

precautionary covariance familiar from the Ramsey literature under aggregate risk (Zhu, 1992; Chari et al., 1994); in this economy it has a first-order object to act on only because the political friction puts one there.

The same decomposition explains the Chilean result. The valuation term is what differs between the two economies: a capital stock that earns several percent a year above depreciation prices the politician's wedge at a rate that moves very little when risk rises, because no ordinary productivity draw brings the return near zero. The political wedge itself is larger in Chile than in the United States, as the low institutional quality of its calibration implies, and it is the division by a large expected return that turns that wedge into a low and stable tax rate.

The limit with no political friction closes the argument: without the politician there is neither a capital tax nor a response of it to risk. As θ approaches one the political wedge vanishes, and what remains is the inert implementability base. At $\theta = 0.95$ the entire fall of the capital tax across this volatility range is 0.010, an order of magnitude smaller than at the calibrated θ .

Through the lens of the model, the volatility patterns of the empirical section are therefore evidence about political frictions. In the standard Ramsey benchmark that the high- θ limit approximates, aggregate risk of empirical size leaves capital taxation essentially unchanged,

consistent with what Zhu (1992) and Chari et al. (1994) find in economies without political frictions. Aggregate risk shapes capital taxation because, and only where, a self-interested politician sustains a capital tax for risk to erode.

6 Conclusions

This paper asks what determines how heavily a country taxes capital, and gives an answer built on aggregate risk. Empirically, more volatile economies tax capital more lightly, the effect is stronger in countries with better institutions, and it does not appear for the labor tax. To rationalize these facts we develop a model in which a self-interested politician sets taxes under aggregate risk, and calibrate it to two reference economies. In the United States aggregate risk lowers the capital tax by 21 percentage points, the reduction is larger where institutions are stronger, and the labor tax does not respond. In Chile, the emerging economy of the pair, the same risk lowers the capital tax by less than a tenth of a percentage point, which is what the data show for countries with weak institutions.

Our results carry one broader message: the effect of aggregate risk on capital taxation is informative about political frictions. In an economy without a politician, risk of the size observed in the data leaves capital taxation essentially unchanged; in our model, the same risk removes 21 percentage points of the politically sustained tax in the advanced reference economy. The two calibrations also show where the mechanism stops: the precautionary motive is present in both economies and shows up in both as a buffer stock of capital, but it reaches the capital tax only where capital barely earns more than it depreciates. A natural next step is to take the model to the full cross-section of countries, letting institutions, volatility and the return to capital vary jointly as they do in the data; we leave this for future research.

A Data sources by country

This appendix records, for each of the 38 economies in the capital-tax regression sample, the years over which the effective tax rates are averaged and the source of every input. All series were retrieved programmatically from the providers' data services rather than transcribed from publications: tax revenue, national accounts and statutory rates from the OECD's SDMX dataflows, the governance indicators from the World Bank Indicators API, the national accounts of the non-OECD economies from the United Nations data portal, and the V-Dem indices from the version 16 country-year file. Missing cells are left empty; nothing is interpolated or back-filled.

OECD members. For the 29 OECD members the sources are uniform. Tax revenue by heading comes from the OECD Revenue Statistics (dataflow `DSD_REV_OECD@DF_REV_ALL`). The national-accounts inputs of the Mendoza et al. (1994) formulas—private and government final consumption, government compensation of employees, wages and salaries, property income, operating surplus of the total economy and of corporations, mixed income, and consumption of fixed capital—come from the OECD Annual National Accounts (tables T0102 and T0800). The labour base is wages and salaries (D.11). Real GDP is the OECD chain-linked volume series, reference year 2020. Statutory corporate rates come from Table II.1 of the OECD Tax Database.

Non-OECD economies. For the nine non-OECD economies, tax revenue by heading comes from the OECD's regional Revenue Statistics dataflows, which apply the same classification as the OECD edition. The national-accounts inputs come from the OECD Annual National Accounts for Bulgaria, Croatia and Romania, with wages and salaries (D.11) as the labour base, and from the United Nations national-accounts institutional-sector datamart for the Dominican Republic, Egypt, Honduras, Kazakhstan, Mongolia and Nicaragua, where the labour base is compensation of employees (D.1). Real GDP is the United Nations Main Aggregates constant-price series, base year 2015.

Economies drawing on two sources. For Chile, Costa Rica and Mexico an input is taken from the OECD source where available and from a secondary source only for the cells the OECD source leaves empty, applied cell by cell: the United Nations national accounts for Costa Rica and Mexico, and the Banco Central de Chile for Chile over 2003–2023. Consumption of fixed capital is not reported for Chile after 2002, so its net capital tax rate,

the one used in the regressions, is available for 1996–2002 only, while the gross rate extends to 2023.

Corrections. The OECD revenue feed labels Chile and Hungary in billions while the values are in millions; the affected rows were rescaled before the rates were computed. Euro-area economies are carried in euros throughout, including before 1999. Table 7 lists the years used and the sources for each economy.

Table 7: Sample coverage and sources by country

Country	ISO	Years averaged		Tax	National accounts	Real
		Capital tax	Labor tax	revenue	(labor base)	GDP
<i>OECD members</i>						
Austria	AUT	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Belgium	BEL	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Chile	CHL	1996–2002	1996–2023	OECD	OECD, BCCh (D.11)	OECD
Costa Rica	CRI	2012–2021	2012–2021	OECD	OECD, UN (D.11)	OECD
Czechia	CZE	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Denmark	DNK	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Estonia	EST	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Finland	FIN	1975–2024	1975–2024	OECD	OECD (D.11)	OECD
France	FRA	1978–2024	1965–2024	OECD	OECD (D.11)	OECD
Germany	DEU	2012–2024	2012–2024	OECD	OECD (D.11)	OECD
Greece	GRC	1995–2023	1995–2023	OECD	OECD (D.11)	OECD
Hungary	HUN	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Iceland	ISL	2000–2024	–	OECD	OECD (D.11)	OECD
Ireland	IRL	2010–2024	2010–2024	OECD	OECD (D.11)	OECD
Italy	ITA	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Japan	JPN	1994–2024	1994–2023	OECD	OECD (D.11)	OECD
Latvia	LVA	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Lithuania	LTU	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Luxembourg	LUX	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Mexico	MEX	2002–2023	2002–2023	OECD	OECD, UN (D.11)	OECD
Netherlands	NLD	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Poland	POL	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Portugal	PRT	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Slovak Republic	SVK	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Slovenia	SVN	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Spain	ESP	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
Sweden	SWE	1965–2024	1965–2024	OECD	OECD (D.11)	OECD
United Kingdom	GBR	1995–2024	1995–2024	OECD	OECD (D.11)	OECD
United States	USA	1992–2024	1992–2024	OECD	OECD (D.11)	OECD
<i>Non-OECD economies</i>						
Bulgaria	BGR	1995–2022	1995–2022	OECD reg.	OECD (D.11)	UN
Croatia	HRV	1995–2024	1995–2024	OECD reg.	OECD (D.11)	UN
Dominican Republic	DOM	2007–2016	2007–2016	OECD reg.	UN (D.1)	UN
Egypt	EGY	2002–2015	2002–2015	OECD reg.	UN (D.1)	UN
Honduras	HND	2000–2019	2000–2019	OECD reg.	UN (D.1)	UN
Kazakhstan	KAZ	1998–2021	1998–2021	OECD reg.	UN (D.1)	UN
Mongolia	MNG	2006–2023	2006–2023	OECD reg.	UN (D.1)	UN
Nicaragua	NIC	2006–2019	2006–2019	OECD reg.	UN (D.1)	UN
Romania	ROU	1995–2023	1995–2023	OECD reg.	OECD (D.11)	UN

Notes. Years averaged: first and last year with a non-missing rate; the country mean uses every available year in between. Tax revenue: OECD = OECD Revenue Statistics; OECD reg. = the OECD’s regional Revenue Statistics dataflows. National accounts: OECD = OECD Annual National Accounts; UN = United Nations national-accounts institutional-sector datamart; BCCh = Banco Central de Chile, used only where the OECD source is empty; D.11 = wages and salaries, D.1 = compensation of employees as the labour base. Real GDP: OECD = chain-linked volume, reference year 2020; UN = Main Aggregates constant prices, base year 2015. Iceland has no labour-tax inputs.

B Derivation of the implementability constraint

This appendix consolidates the household's optimality and budget conditions into the single intertemporal constraint (15), following the primal approach of Lucas and Stokey (1983).

The household chooses $\{c_t(s^t), n_t(s^t), k_{t+1}(s^t), b_{t+1}(s_{t+1} | s^t)\}$ to maximize its expected utility subject to the flow budget constraint of the Households subsection. Writing $\pi_t(s^t)$ for the probability of history s^t , its first-order conditions deliver the intratemporal labor condition

$$-\frac{u_n(c_t(s^t), n_t(s^t))}{u_c(c_t(s^t), n_t(s^t))} = (1 - \tau_{\ell,t}(s^t)) w_t(s^t), \quad (29)$$

and the no-arbitrage condition that prices the state-contingent claims,

$$p(s_{t+1} | s^t) = \beta \pi(s_{t+1} | s^t) \frac{u_c(c_{t+1}(s^{t+1}), n_{t+1}(s^{t+1}))}{u_c(c_t(s^t), n_t(s^t))}. \quad (30)$$

The household's Euler equation for capital prices the return on capital consistently with these claims, so that there is no arbitrage between capital and bonds.

Define the cumulative price of history s^t ,

$$q_t(s^t) \equiv \prod_{i=1}^t p(s_i | s^{i-1}) = \beta^t \pi_t(s^t) \frac{u_c(c_t(s^t), n_t(s^t))}{u_c(c_0, n_0)}, \quad (31)$$

where the second equality iterates (30). Multiplying each flow budget constraint by $q_t(s^t)$ and summing over all dates and histories, the capital and bond positions telescope under the transversality conditions, leaving the present-value budget constraint

$$\sum_{t=0}^{\infty} \sum_{s^t} q_t(s^t) \left[c_t(s^t) - (1 - \tau_{\ell,t}(s^t)) w_t(s^t) n_t(s^t) \right] = [1 + (1 - \tau_{k,0})(r_0 - \eta)] k_0 + b_0. \quad (32)$$

Substituting the labor condition (29) and the price (31) into (32) and multiplying through by $u_c(c_0, n_0)$ yields

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t) [u_c(c_t(s^t), n_t(s^t)) c_t(s^t) + u_n(c_t(s^t), n_t(s^t)) n_t(s^t)] = u_c(c_0, n_0) \{ [1 + (1 - \tau_{k,0})(r_0 - \eta)] k_0 + b_0 \},$$

which is the implementability constraint (15), since $\mathbb{E}_0[\cdot] = \sum_{s^t} \pi_t(s^t)[\cdot]$. $\square$

C Proof of Lemma 1

The argument has two steps. The first characterizes the best sustainable equilibrium as a constrained welfare-maximization problem; the second verifies that this characterization is undisturbed by the joint presence of physical capital and aggregate productivity risk.

Step 1 (sustainability). Because the politician cannot commit, a policy is part of a subgame-perfect equilibrium if and only if, at every date and history, the politician does not gain from a one-shot deviation (the one-shot-deviation principle). Because v is strictly increasing, the politician gains most by appropriating the entire divertible share, so the most profitable deviation is to seize a fraction $1 - \theta$ of current output and then be replaced; this yields $v((1 - \theta)Y_t(s^t))$ followed by the normalized punishment value $v(0) = 0$ in every subsequent period—the harshest credible punishment, in the sense of the optimal penal code of Abreu (1988). Comparing this payoff with the continuation value of compliance delivers the Political Participation Constraint (12) as the necessary and sufficient condition for sustainability. The best sustainable equilibrium is therefore the competitive equilibrium that maximizes the household’s welfare subject to (12).

Step 2 (capital and aggregate risk). Acemoglu et al. (2008) establish this characterization in a deterministic economy with physical capital, including the convexification of the non-convex sustainability set by a public-randomization device, under which the value function is concave and the characterization is exact (the randomization is degenerate at the optimum). Introducing the exogenous, finite-state Markov productivity process enters the problem only through the conditional-expectation operator $\mathbb{E}_t[\cdot]$, which is linear and hence preserves the concavity of the objective—concave because the period utility u is concave—and the convexity of the (lottery-extended) constraint set; moreover the deviation payoff $v((1 - \theta)Y_t(s^t))$ depends only on the current state. The deterministic characterization therefore applies history by history, and the treatment of aggregate risk in Yared (2010) carries over to the present environment with capital. $\square$

D From the planning problem to the recursive saddle point

This appendix derives the recursive saddle-point problem (16)–(18) from the planning problem (14), following the recursive-contracts method of Marcet and Marimon (2019). The

difficulty is that the Political Participation Constraint (12) is forward looking—it involves a discounted sum of future rents—so problem (14) does not admit a Bellman representation in the physical state alone: the current choice of appropriation carries an implicit promise about future appropriation. The remedy is to dualize the forward-looking constraint and let its accumulated multiplier serve as an additional state.

Step 1 (Lagrangian of (P)). Attach a single time-invariant multiplier ψ to the intertemporal implementability constraint (15) and, at each date and state, a multiplier $\mu_t \geq 0$ to the PPC (12). Substituting the resource constraint (13) to eliminate g where convenient and keeping it as an explicit constraint otherwise, the Lagrangian of (P) is

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, n_t) - \psi [u_c(c_t, n_t) c_t + u_n(c_t, n_t) n_t] \right\} + \psi \Omega_0 + \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \left[\mathbb{E}_t \sum_{s=0}^{\infty} \delta^s v(g_{t+s}) - v((1-\theta)A_t F(k_t, n_t)) \right],$$

where $\Omega_0 \equiv u_c(c_0, n_0) \{ [1 + (1 - \tau_{k,0})(r_0 - \eta)]k_0 + b_0 \}$ is the (constant) right-hand side of (15) and the resource constraint (13) is adjoined with its own multiplier, suppressed here for readability. The single multiplier ψ reflects that (15) is one intertemporal constraint, fixed at $t = 0$; the multipliers μ_t are date- and state-specific because the PPC must hold at every node.

Step 2 (folding the forward-looking sum). The non-recursive object is the inner term $\mathbb{E}_t \sum_{s=0}^{\infty} \delta^s v(g_{t+s})$. Consider its contribution to $\mathcal{L}$,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \mathbb{E}_t \sum_{s=0}^{\infty} \delta^s v(g_{t+s}).$$

By the law of iterated expectations, $\mathbb{E}_0[\beta^t \mu_t \mathbb{E}_t[\cdot]] = \mathbb{E}_0[\beta^t \mu_t (\cdot)]$, so we may exchange the order of summation and collect, for each calendar date τ , every term in which $v(g_\tau)$ appears. The rent $v(g_\tau)$ is multiplied by $\beta^t \mu_t \delta^s$ for every pair (t, s) with $t + s = \tau$ and $t \leq \tau$, that is by $\sum_{t=0}^{\tau} \beta^t \mu_t \delta^{\tau-t}$. Hence

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \mathbb{E}_t \sum_{s=0}^{\infty} \delta^s v(g_{t+s}) = \mathbb{E}_0 \sum_{\tau=0}^{\infty} \left(\sum_{t=0}^{\tau} \beta^t \mu_t \delta^{\tau-t} \right) v(g_\tau) = \mathbb{E}_0 \sum_{\tau=0}^{\infty} \beta^\tau \gamma'_\tau v(g_\tau),$$

where the last equality defines the co-state

$$\gamma'_\tau \equiv \beta^{-\tau} \sum_{t=0}^{\tau} \beta^t \mu_t \delta^{\tau-t} = \sum_{t=0}^{\tau} \left(\frac{\delta}{\beta} \right)^{\tau-t} \mu_t.$$

This is precisely the device of Marcet and Marimon (2019): the discounted future terms of

the forward-looking constraint are folded back into the period return by replacing the period weight on $v(g_\tau)$ with the accumulated multiplier γ'_τ .

Step 3 (co-state law of motion). The definition of γ'_τ implies the recursion

$$\gamma'_\tau = \frac{\delta}{\beta} \gamma'_{\tau-1} + \mu_\tau, \quad \gamma'_{-1} = 0,$$

which, dropping the calendar index and writing $\gamma \equiv \gamma'_{\tau-1}$ for the entering co-state and $\gamma' \equiv \gamma'_\tau$ for the updated one, is the law of motion (18),

$$\gamma' = \frac{\delta}{\beta} \gamma + \mu.$$

The co-state thus accumulates the past PPC multipliers, discounted by the ratio δ/β of the political to the household discount factor; the initialization $\gamma_0 = 0$ corresponds to no inherited promise at the start of the program.

Step 4 (time-additive Lagrangian and the saddle point). Collecting Steps 1–3, and using that the deviation payoff $v((1-\theta)A_t F(k_t, n_t))$ depends only on the current state, the Lagrangian becomes a discounted sum of time-invariant return functions of the augmented state,

$$\mathcal{L} = \psi \Omega_0 + \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, n_t) + \gamma'_t v(g_t) - \mu_t v((1-\theta)A_t F(k_t, n_t)) - \psi [u_c(c_t, n_t) c_t + u_n(c_t, n_t) n_t] \right\},$$

subject to the resource constraint (13) and the co-state law of motion (18). Because the constant $\psi \Omega_0$ does not affect the optimization and the productivity process is Markov, the problem now has (k, γ, A) as its state and a return that is the same function at every date. The planner maximizes over the allocation (c, n, g, k') and minimizes over the PPC multiplier $\mu \geq 0$, so the value function $V(k, \gamma, A)$ satisfies the saddle-point functional equation

$$V(k, \gamma, A) = \min_{\mu \geq 0} \max_{c, n, g, k'} \left\{ u(c, n) + \gamma' v(g) - \mu v((1-\theta)A F(k, n)) - \psi [u_c(c, n) c + u_n(c, n) n] + \beta \mathbb{E}[V(k', \gamma', A')] \right\},$$

subject to (17) and (18). This is equation (16). By Theorems 1 and 2 of Marcet and Marimon (2019), the solution of this saddle-point functional equation reproduces the solution of the original planning problem (14), with γ' equal to the accumulated multipliers on the Political Participation Constraint and the continuation problem from the updated co-state γ' reproducing the continuation of the original solution.

Step 5 (first-order conditions). Differentiating the saddle-point objective and using the

envelope condition $V_k(k, \gamma, A) = (u_c(c, n) - \psi \Gamma)(1 - \eta + AF_k(k, n)) - \mu(1 - \theta)v_g((1 - \theta)AF(k, n))AF_k(k, n)$ together with $\partial\gamma'/\partial\mu = 1$, the optimality conditions with respect to g, c, n , and k' deliver (19)–(21), where $\Gamma \equiv u_{cc}c + u_c$ and $\Lambda \equiv u_{nn}n + u_n$, and the minimization over $\mu \geq 0$ delivers the recursive PPC (22) with complementary slackness. $\square$

E Proofs for Section 4

Proof of Lemma 2. (i) Stationarity of the co-state law $\gamma' = (\delta/\beta)\gamma + \mu$ gives $\mu^{SS} = (1 - \delta/\beta)\gamma^{SS}$. Suppose $\gamma^{SS} = 0$. Then the planner attaches no weight to rents, so $g^{SS} = 0$ (rents cost resources and enter the objective only through $\gamma'v(g)$), and the value of complying is $v(0)/(1 - \delta) = 0$; but for $\theta < 1$ the deviation payoff $v((1 - \theta)y^{SS})$ is strictly positive, so the Political Participation Constraint is violated. Hence $\gamma^{SS} > 0$ and, with $\delta < \beta$, $\mu^{SS} > 0$. (ii) At a steady state the household's Euler equation with a constant capital tax reads $1 = \beta[1 + (1 - \tau_k^{SS})(r^{SS} - \eta)]$, so $(1 - \tau_k^{SS})(r^{SS} - \eta) = 1/\beta - 1$ and therefore $(1 - \eta + r^{SS}) - 1/\beta = \tau_k^{SS}(r^{SS} - \eta)$. Substituting into the implementability term of the wedge (which at a steady state, where $\Gamma' = \Gamma$, equals $\psi \Gamma[(1 - \eta + r) - 1/\beta]$) and dividing by the pricing term $u_c(r - \eta)$ gives $\bar{\tau}_k^{imp, SS} = \psi(\Gamma/u_c)\tau_k^{SS} = \psi(1 - \sigma_c)\tau_k^{SS}$. (iii) By (ii), $[1 + \psi(\sigma_c - 1)]\tau_k^{SS} = \bar{\tau}_k^{pol, SS} = \mu^{SS}(1 - \theta)v_g((1 - \theta)y^{SS})r^{SS}/[u_c^{SS}(r^{SS} - \eta)]$. With $v_g(x) = x^{\nu-1}$ the political numerator is proportional to $(1 - \theta)^\nu$, which vanishes as $\theta \rightarrow 1$ while the remaining steady-state objects converge to those of the frictionless Ramsey steady state; since $1 + \psi(\sigma_c - 1) > 0$ (it equals $\gamma'v_g(g)/u_c$ at the steady state, positive by (i)), $\tau_k^{SS} \rightarrow 0$. As $\delta \rightarrow \beta$, $\mu^{SS} = (1 - \delta/\beta)\gamma^{SS} \rightarrow 0$ with γ^{SS} bounded, so the same expression gives $\tau_k^{SS} \rightarrow 0$. $\square$

Proof of Proposition 1. (i) If $\gamma_{t+1} = \gamma_t = \gamma^*$ then the co-state law gives $\mu_t = (1 - \delta/\beta)\gamma^*$ at every t ; conversely a constant μ^* makes γ converge to and stay at $\mu^*/(1 - \delta/\beta)$, and constancy from date zero requires exactly $\mu_t = (1 - \delta/\beta)\gamma_t$. (ii) Immediate from the co-state law with $\mu_t = 0$ and $\delta < \beta$. (iii) Take expectations of the co-state law under the stationary distribution: $\mathbb{E}[\gamma] = (\delta/\beta)\mathbb{E}[\gamma] + \mathbb{E}[\mu]$. If the constraint binds with positive probability then $\mathbb{E}[\mu] > 0$. $\square$

Proof of Proposition 2. (i) The wedge Ξ' is the sum of its political and implementability terms (the Taxes subsection), and the conditional expectation is linear, so dividing by the common denominator splits $\bar{\tau}_k(z)$ exactly. (ii) By inspection of the two terms: θ appears in $\mu'(1 - \theta)v_g((1 - \theta)y')r'$ and nowhere in $\psi[\Gamma'(1 - \eta + r') - \Gamma/\beta]$. (iii) With $\sigma_c > 1$, $\Gamma = (1 - \sigma_c)u_c(c)$ is strictly increasing in c ; with Cobb–Douglas technology, $r' = \alpha y'/k'$ at the

chosen k' , so r' is a strictly increasing transformation of y' . The hypothesis therefore makes Γ' and r' comonotone across successor states, and for any two comonotone random variables on a finite probability space the covariance is nonnegative, by the discrete Chebyshev inequality $\text{Cov}(X, Y) = \frac{1}{2} \sum_{s,t} \pi_s \pi_t (X_s - X_t)(Y_s - Y_t) \geq 0$. Multiplying by $\psi < 0$ gives the sign. At $\sigma = 0$ the conditional distribution is degenerate and the covariance is zero. $\square$

Proof of Corollary 1. Immediate from Proposition 2(ii) and (iii): the implementability component contains no θ and no risk-signed term, so any dependence of the tax on institutional quality, and any signed effect of risk on the pricing, operate through the political component. $\square$

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